The Information Content of CEOs' Personal Social Media: Evidence from Stock Returns and Earnings Surprise

Xing Gao*

University of Illinois at Urbana-Champaign

May 2019

Abstract

We study the personal Twitter accounts of 226 CEOs, and examine whether the content across 127,916 Tweets hold information about future firm performance. The content of these Tweets mostly deals with CEOs' personal activities and opinions, and the analysis further employs machine learning algorithms to identify and exclude Tweets that have relevance for firm operations. The results show that a high proportion of positive words in the CEOs' Tweets predicts positive future abnormal returns. This pattern is especially strong before earnings release. Using Tweets posted before earnings announcements, a daily calendar-time spread portfolio that buys stocks with a high proportion of positive words and sells stocks with a low proportion delivers a return of 10.8 to 19.6 basis points per day. Furthermore, the cumulative proportion of positive words can predict quarterly earnings surprise, earnings and cash flows. These results suggest that the content of personal Tweets can elicit the moods of CEOs, and that executives tend to spend time doing enjoyable activities when they have achieved the performance targets.

JEL category: G14, G34

Keywords: Information Content, Social Media, CEOs' Tweets

*Xing Gao is at the University of Illinois at Urbana-Champaign; Gies College of Business; 1206 South Sixth Street; Champaign, Urbana, IL 61820; U.S.A.; E-mail: hgao12@illinois.edu.

Electronic copy available at: https://ssrn.com/abstract=3293134

Introduction

Research has examined important information sources about firms' fundamentals: analyst forecasts, accounting variables of financial statements, and media coverage of firms, such as in The Wall Street Journal and Dow Jones News Services. This paper looks at an alternative data source: Chief Executive Officers (CEO) personal social media. We use the Tweets of CEOs made during their tenure, and find a close connection between the proportion of positive words in their tweets and firms' fundamentals. Unlike public social media, which reports firm operations and prospects, CEOs tweet their personal life activities and opinions using their personal Twitter accounts.

We classify words with positive or negative sentiment using the natural language toolkit, which contains 2006 positive words and 4783 negative words, to capture the information in Tweets. For example, "A wonderful evening incredible food merci lachevredor" contains 2 positive words ("wonderful" and "incredible") out of 7 words. Another example, "What a gorgeous Saturday morning in Miami. May jump to the Caribbean one day this week. Barbados has a nice ring to it" includes 2 positive words ("gorgeous" and "nice"). Finally the example "DC today, NYC tonight and back to DC tomorrow. Crazy schedule with lots of good things going on" has only one positive word ("good"). We use the proportion of positive words as a proxy for information about firms. If the fraction of positive words in Tweets is a noise measurement of positive information about the firm, the coefficient of positive words in the regressions on firms' fundamentals should be zero. Similarly, if the proportion of positive words in Tweets is a proxy for positive information about firms' fundamentals that is already incorporated into stock prices, we should find that it has no predictability for future market activities.

We find a positive correlation between the proportion of positive words in the Tweets of CEOs and firms' stock returns. That is, the higher the proportion of positive words in a Tweet, the higher the daily stock return. This is particularly shown to occur before earnings announcements. The positive correlation can be driven by three competing hypotheses: sentiment, conspiracy and information. First, sentiment theory believes that irrational investors misinterpret the positive words in the Tweets and buy stocks, which

Electronic copy available at: https://ssrn.com/abstract=3293134

causes a stationary increase in returns. Thus, returns rebound during the next period when there is a new belief about the stock. Second, believers in the conspiracy hypothesis believe that a temporary increase in the stock price is caused by false information due to CEO corruption. Thus, stock prices will fall when investors learn the true information. Third, information theory supports the contention that the positive words in Tweets are proxies for firms' fundamentals that are not currently incorporated into stock prices. We examine the cumulative abnormal return around earnings announcements and find that the cumulative abnormal return increases without a reversal around the day of the Tweet, which supports the information theory.

We also find that positive words in Tweets posted before earnings announcements can predict subsequent stock returns. This is because CEOs already know whether their firms' earnings have beaten analysts' forecasts, while investors are not informed of such information. Compared to CEOs who have failed to meet the analysts forecasts, CEOs who have achieved the performance target are more likely to be happier and engage in enjoyable activities, which are reflected in Tweets as positive words. Therefore, we can use CEO Tweets to infer the mood of CEOs before earnings release. Consequently, we construct daily calendar-time spread portfolios that buy stocks with a high proportion of positive words in their CEOs Tweets and sell stocks with a low fraction of positive words in Tweets posted during 14 days prior to earnings announcements. The portfolio of stocks with positive words in the top 10th and the portfolio with stocks in the bottom 10th percentile have an alpha of 12.8 and -4.4 basis points, respectively. A spread portfolio that buys stocks with the proportion of positive words in the top 10th percentile and sells stocks in the bottom 10th percentile has an alpha of 17.2 basis points per day.

We further show that the proportion of positive words has predictive power for firms' earnings. That is, the cumulative fraction of positive words can predict the difference between actual earnings and analysts' forecasts, which is known as "earnings surprises." These findings are consistent with the information hypothesis. That is, when CEOs are confident and satisfied with firm performance, they tend to spend time doing enjoyable activities and are in a pleasant mood. For example, they may spend time with family and friends, and tweet about dining out or vacations. Or, they may simply be happier and

Electronic copy available at: https://ssrn.com/abstract=3293134

enjoy the weather or scenery. Thus, they use more positive words in their Tweets, such as "gorgeous, incredible, good, happy, nice, terrific, amazing, wonderful and beautiful."

We next examine whether the content of Tweets is an unintentional informational release without an agenda, or a case of CEOs strategically choosing the time to disclose the information in order to maximize their personal payoff. We compare the characteristics, including the proportion of positive words, in various Tweets posted around the time of CEO option exercises and stock sales. In a manner consistent with the timing hypothesis, the Tweets of CEOs contain a higher fraction of positive words before, as opposed to after option exercises and stock sales.

In a nutshell, by using a novel dataset about CEO personal activities and opinions, we find that the positive words in the personal Tweets of CEOs can predict daily stock returns and quarterly earnings. In addition, the Tweets of CEOs contain a higher proportion of positive words when they are incentivized to increase the stock price, such as before option exercises and stock sales. Together these actions show that CEOs strategically time the release of Tweets, which contain information about the firms' fundamentals. These findings can be used to form trading strategies. For example, long stocks that belong to firms which have positive words in their CEO Tweets, while shorting the stocks which lack positive words.

The paper is structured as follows. Section 1 presents research literature on information content and earnings manipulation. It also reviews papers using text data from news stories. Section 2 will discuss the methodology, data sources and variable constructions used in the study. Section 3 and 4 will present the main results on predicting stock returns and earnings using positive words in Tweets, respectively. Section 5 explores CEO motivation to utilize positive words. Section 6 concludes the paper.

1 Related Literature

Stories from financial media provide qualitative information that can be used to study the effect on market activities. Matching news stories from The Wall Street Journal and Dow Jones News Services to firms, Tetlock, Saar-Tsechansky, and Macskassy (2008) show that

Electronic copy available at: https://ssrn.com/abstract=3293134

the negative words in the financial press have predictive power regarding firms' earnings and stock returns. Thus, they show that there is information content in the linguistic descriptions in social media about firm operations, which is not yet incorporated into quantitative accounting variables. In contrast to the information channel through which stories in financial press affect investors, Tetlock (2007) finds that negative words in the "Abreast of the Market" column of The Wall Street Journal (WSJ) influence investors though sentiments. That is, media pessimism, measured by the negative words in WSJ, can predict lower market returns. Since the author finds that temporarily low returns are reversed shortly after news stories releases, he interprets this temporary effect on market as merely sentiments. Antweiler and Frank (2006) use the computational linguistics methods to classify topics of corporate stories from WSJ. They find that the market overreacts to news information, and returns reverse after 10 days of the news event.

Financial statements, earnings announcements and messages from internet chat rooms are other important sources of qualitative text information. Li (2006) measures the "risk sentiment" in annual statements by counting the frequency of "risk" and "uncertain". He finds that an increase in "risk sentiment" can predict lower earnings. Davis, Piger, Sedor et al. (2006) use a dictionary-based content analysis program to measure the level of optimistic and pessimistic language in earnings press release. They show a correlation between the level of optimistic language usage in earnings announcements and firms' future performance. Messages from internet chat rooms are used to study trading patterns. Antweiler and Frank (2004) identify "sell", "buy" and "hold" activities from internet stock message board, which predict market volatilities.

In this study, we use the personal social media of CEOs as a source of information about firms, which is different than the previous literature that uses corporate news releases, financial press stories, and financial statements. Firms and CEOs have separate Twitter accounts: the former concerns about firm operations, and business prospects, similar to corporate news releases, and the latter concerns CEOs' personal activities and opinions. These personal Tweets cover topics that include vacation, food, travelling, time spent with family and friends, games, weather, scenery and so on. A small proportion of them concerns firm activities, and we try to exclude them. To be more specific, in a thousand

Electronic copy available at: https://ssrn.com/abstract=3293134

randomly selected Tweets of CEOs, we identify twenty-seven firm-related Tweets. We then use machine learning algorithms to identify and exclude these Tweets which are related to firm operations.

This paper relates to the information content of CEOs' activities. Tracking corporate jet flights, Yermack (2014) finds a connection between CEO vacation travelling and the timing of corporate information releases. He finds that there is an information content regarding CEO absences from headquarters, where corporate news will be withheld and released right after CEOs are back. Similarly, we identify the information content of CEOs' activities by tracking their Tweets. Intuitively, when CEOs are confident and satisfied with their firms' performance, they tend to feel happier or engage in enjoyable activities. For example, they use positive tones to describe weather or scenery, and positive words to express the activities they engage in when they spend with family to dine out. Therefore, the positive words in CEOs' Tweets can predict firms' future performance.

This study contributes to literature related to managers' manipulation of the timing of corporate voluntary disclosures. CEOs are incentivized to lower the stock price on option grant dates, because option exercise price is set to be the stock price on grant days. Aboody and Kasznik (2000) find that CEOs manage to quickly release bad news and delay announcements of good news about earnings before option grant dates to increase the fair value of executives' compensation. In a manner that is consistent with this reasoning, Chauvin and Shenoy (2001) show that the stock prices prior to stock option grants tend to decrease due to the release of bad discretionary news about earnings, dividends and forecasts. Yermack (1997) examines stock price movements after option awards and finds that price increases following option grants are caused by favorable earnings announcements.

In this paper, we find another informal communication channel through which executives transmit information to investors, which is the personal social media of CEOs. As Chauvin and Shenoy (2001) point out, executives do not want to spoil their compensation packages before compensation committee meetings by formally releasing bad news. Instead, executives manage to manipulate stock prices downward using informal or private

Electronic copy available at: https://ssrn.com/abstract=3293134

communication methods such as rumors and innuendo, which can be attributed to market misinformation.

In addition to the timing of the release of corporate news, there are other methods executives can use to manipulate and maximize their compensation. Baker, Collins, and Reitenga (2003) show that managers use income-decreasing discretionary accruals to lower stock prices prior to option grants. Managers are also incentivized to manipulate stock prices upward, leading up to option exercises. Balachandran, Chalmers, and Haman (2008) suggest that firms use buyback programs to maximize option payoff. However, this positive correlation between share repurchases and exercisable options can be driven by confounding variables such as firm life cycle or investment opportunities. Bartov and Mohanram (2004) show that executives use private information about positive earnings to time abnormally large option exercises.

Finally, this study is related to research which uses big data and machine learning. Huang (2018) examines 14 million customer product reviews on Amazon.com, and documents the information content of consumer opinions concerning stock prices. Luca (2011) similarly examines customer review ratings on Yelp.com and shows a causal impact on revenue using a framework of regression discontinuity. In terms of machine learning, Erel, Stern, Tan, and Weisbach (2018) develop algorithms to select board of directors based on firm characteristics and director features. They claim that compared to ordinary least square regressions, machine learning techniques achieve more accurate out-of-sample predictions. In this paper, we use a linear stochastic gradient descent classifier to identify and remove Tweets related to firm operations.

2 Methodology and Data

We construct the sample by manually identifying the personal Twitter accounts of CEOs at Twitter.com. We start with a list of three thousand CEOs who are in position on or after 1/1/2011 using the Annual Compensation dataset from ExecuComp. As Twitter was founded in 2006 and it became more popular after 2011, we exclude CEOs who left their positions before 2011. We identify CEO Twitter accounts using a name search

Electronic copy available at: https://ssrn.com/abstract=3293134

and carefully examine accounts descriptions, profile pictures, the number of followers and Tweets to distinguish the real CEO from people with the same names, and fake or propaganda accounts. For example, when we search for "Tim Cook," the CEO of Apple Inc., we find 46 accounts with the same name, or similar names, including "Timothy Cook". However, the real account is verified by Twitter and followed by 11 million people. In the description, it includes "Apple CEO". We find 327 CEO personal accounts, which belong to 335 firms. It is important to note that, we do not collect any Tweets from the firms' official Twitter accounts, which contain information about products, services, operations and prospects. Next, we download the tweets from the personal Twitter accounts of CEOs and remove Tweets that are posted outside of CEO tenure. We use "date became CEO" and "date left as CEO" from ExecuComp to define CEO tenure. If a CEO has served in multiple firms, some Tweets are attributed to previous firms based on the dates he was the CEO. We drop the accounts if all of the Tweets are posted before or after CEO tenure, if they are of a private nature and prohibit me from downloading the Tweets, or if they belong to firms that have assets of less than 1 million dollars. We end up with 226 CEO personal Twitter accounts and 127,916 Tweets during their tenure.

CEOs tweet messages about various topics, such as cooking, dining out, travelling, vacations, doing sports, appreciating arts, kids' progress, hanging out with friends, spending time with family, watching games, attending social rituals, firm operations and so on. This study focuses on the information content of their personal activities. Thus, we exclude Tweets about firm operations. To do this, we first randomly draw 1000 Tweets, which include 500 for training and 500 for testing purposes, and manually identify which of them are related to firm operations. We find 27 Tweets that are related to firm activities from 1000 randomly selected Tweets. It makes sense because firms have other official Twitter accounts to disclose information about firm activities or to promote firm products and services. Next, we design a linear stochastic gradient descent (SGD) classifier algorithm on the training dataset and perform classification on the test dataset. To be more specific, the algorithm breaks the text of a Tweet into tokens, which are used to construct a bag of words. It transforms a text document into a matrix that counts the number of times the words appear in each Tweet. Linear SGD classifier uses a logistic regression

Electronic copy available at: https://ssrn.com/abstract=3293134

with a cost function, specifying the loss of over-fitting. This machine learning algorithm achieves an accuracy rate of 98% on the test dataset. Next, we perform classifications on the remaining one hundred thousand Tweets.

In order to measure the information content of Tweets, we start with cleaning up the Tweets by removing all hash-tags (#MAGA), Twitter handles (@KellyannerPolls), URLs, and punctuations. Natural Language Toolkit contains 2006 positive words and 4783 negative words. For instance, the first ten positive words are "a+, abound, abounds, abundance, abundant, accessible, accessible, acclaim, acclaimed, acclamation." The first ten negative words are "2-faced, 2-faces, abnormal, abolish, abominable, abominable, abominate, abomination, abort and aborted." Next, we count the number of positive words and negative words in each Tweet. For example, a Tweet of "What a gorgeous Saturday morning in Miami. May jump to the Caribbean one day this week. Barbados has a nice ring to it" contains two positive words ("gorgeous" and "nice") and zero negative words. Next, we define the proportion of positive words, the proportion of negative words and the total number of words in each Tweet. In the above example, the fraction of positive words is 0.09, the fraction of negative words is 0, and the total number of words is 21.

Stock price data is drawn from the Center for Research in Security Prices (CRSP). We match daily stock price information with Tweets using the timestamp of each Tweet. If the Tweet is posted when stock market is closed, then the Tweet is matched with the stock information for the next day. We also include the Fama-French three research factors from the Kenneth R. French Data Library. Control variables on firm characteristic and CEO features are collected from the Compustat and ExecuComp databases, respectively. These variables include firm size (log of total assets), cash-to-assets, market-to-book ratio, return-on-assets, market leverage, CEO total compensation, CEO age and CEO tenure (current year minus year became CEO). These control variables are measured at the end of the preceding fiscal year.

Data about earnings surprise comes from the surprise history of summary history of the Institutional Brokers' Estimate System (IBES). We collect variables including date of the actual earnings announcement, actual earnings, and the mean and the standard deviation

Electronic copy available at: https://ssrn.com/abstract=3293134

of analysts' earnings forecasts. Standardized unexpected earnings (SUE) is defined as the difference between the actual earnings and the mean of analysts' earnings forecasts (earnings surprise), divided by the standard deviation of the earnings surprise. We use quarterly frequency, which is the highest frequency of earnings data.

Other measurements of earnings include earnings per share (EPS) and normalized earnings, which are constructed using data drawn from Compustat. Earnings per share are measured in various ways: diluted EPS divided by lagged stock price, basic EPS divided by lagged stock price, income before extraordinary items divided by lagged capitalization, income before extraordinary items adjusted for common stock equivalents divided by lagged capitalization, and income before extraordinary items available for common divided by lagged capitalization. For normalized earnings, we define them as the following alternatives: four times net income divided by lagged assets, income before extraordinary items divided by lagged assets, income before extraordinary items adjusted for common stock equivalents divided by lagged assets, income before extraordinary items available for common divided by lagged assets, and cash flow (sum of net income and depreciation expenses) divided by lagged assets. These outcome variables on earnings are measured at the end of each quarter.

Data about stock option exercises and stock sales is collected from the Table 2 and Table 1 from the Thomson Reuters database. We use the managers' role titles to restrict the sample of managers to CEOs only. For the option exercises from the Table 2 dataset, we restrict the form type that the insiders file to be 4, the nature of the underlying transaction to be exercise, and the derivative type to be employee stock options, incentive stock options, options, non-qualified stock options, or call options. For stock sales from Table 1 dataset, we limit the form type to be 4, the transaction type to be open market or private sale, and the security title to be common stocks. We collect the dates of CEO option exercises and stocks sales.

[Insert Table 1 about here]

Table 1 reports the summary statistics concerning Tweets in Panel A, and firm and

Electronic copy available at: https://ssrn.com/abstract=3293134

CEO characteristics in Panel B. Among the 127,916 Tweets posted during CEO tenure, the average proportion of positive words is 0.072 and the mean of proportion of negative words is 0.018. The total number of words in a Tweet is 14.19 on average. The average fraction of Tweets related to firm operations is 0.018. The mean of stock returns on Tweet days is 0.01, small minus big factor is 0.012, high minus low factor is -0.017, and market premium is 0.055. Panel B of Table 1 shows firm and CEO characteristics at the end of the preceding fiscal year. The cash to asset ratio is 0.2, and total assets is 20 billion dollars on average. The mean of market-to-book is 5.197, return-on-assets is 0.037 and market leverage is 0.342. The mean of total compensation is 6.8 million dollars, which includes salary, bonuses, other annual sums, the total value of restricted stock grants, the total value of stock option grants, and long-term incentives payouts. The average age of CEOs is 53 years and they have been in the position for 7 years.

3 Using CEOs' Tweets to Predict Stock Returns

3.1 Contemporary Returns

In this section, we test the hypothesis that positive words in the Tweets of CEOs contain information that can predict stock returns. First, we explore the relationship between positive words used in Tweets and daily stock return on the days of Tweets. Next, we use regression analysis to explore the predictive power of the proportion of positive words for stock returns, followed by examining the cumulative abnormal returns around Tweet days in an event study framework.

[Insert Figure 1 about here]

To start with, Figure 1 presents an intuitive relationship between positive words in Tweets and stock returns. Panel A plots daily stock returns after CEOs post Tweets on bins with regards to the proportion of positive words in Tweets. The proportion of positive words is defined as the number of positive number words divided by the total

Electronic copy available at: https://ssrn.com/abstract=3293134

number of words in each Tweet. Bin 0 includes Tweets with zero positive words, bin 1 contains Tweets with a (0, 0.1] proportion of positive words, bins 2 to 4 contain a (0.1, 0.2], (0.2, 0.3] and (0.3, 0.4] proportion, respectively, and bin 5 contains Tweets with more than a 0.4 fraction. The mean of daily returns of CEOs with zero positive words is 3 basis points. The average stock return for firms within bin 1 increases to 5 basis points. The average return in bin 5, which includes Tweets that have more than 40% positive words, is 13 basis points. As the proportion of positive words increases, the stock return also increases. Panel B plots the daily stock return on the number of positive words. The bars from left to right correspond to Tweets that have 0, 1, 2, 3, 4, and 5 or more positive words. If a Tweet contains more than 5 positive words, it is included in the last bar with 5 positive words. Similar to Panel A, as the number of positive words increases, the stock return also increases.

There are competing theories explaining the positive relationship between positive words in Tweets and stock returns. First, sentiment theory believes that irrational investors misinterpret the positive words in Tweets and they buy stocks that will cause a stationary increase in returns. Thus, based on sentiment theory, returns will rebound in the next period when there is a new belief about the firm in the market. Second, the conspiracy hypothesis holds that a temporary increase in stock price is caused by false information due to CEO corruption. Thus, conspiracy theory predicts that stock prices will fall after investors discover the truth. Third, information theory supports the contention that positive words in Tweets are proxies about firms' fundamentals that are not currently incorporated into stock prices. Therefore, information theory predicts that an increase in returns will persist indefinitely.

[Insert Table 2 about here]

To determine whether CEOs' Tweets can predict stock returns, we use regression analysis. Using 0.12 million Tweets of CEOs, Table 2 explores the predictive power of the proportion of positive words for daily stock returns using the following regression

Electronic copy available at: https://ssrn.com/abstract=3293134

equation:

$$\text{Daily Stock Return} = \alpha + \beta_1 \text{Pos. Prop.} + \beta_2 \text{Neg. Prop.} + \beta_3 \text{Tot. Words} + \text{Controls}$$

The dependent variable is the stock return on Tweet days. Note that if a Tweet is posted when the market is closed, the stock return on Tweet days is replaced with that of the next day. We control for other features of Tweets, including the proportion of negative words and the total number of words in a Tweet. We also include Fama-French three factors to capture market conditions. As regards firm characteristics, we include firm size, cash-to-assets, market-to-book ratio, return-on-assets, and market leverage. CEO features include total compensation, age and tenure. We control for the industry or firm interacting with year and quarter fixed effects in columns 1 or 3, and the industry or firm interacting with year and month fixed effect in columns 2 or 4. Panel A excludes Tweets related to firm operations. The coefficient of the proportion of positive words is 0.0013 on the daily stock return, and it is significant at the 5% level. This means that with a 1% increase in the fraction of positive words, the daily return will increase by 0.13 basis points. Given that the average proportion of positive words is 10%, posting a Tweet with 10% positive words will increase the daily stock return by 1.3 basis points. The magnitude does not change significantly when we include firm and month fixed effects. In Panel B, we use all of the Tweets of CEOs, which include Tweets related to firm operations and activities. The coefficients of the proportion of positive words increase to 0.0014 and are significant at the 5% level of significance.

As a robust check, which mitigates the concern of reverse causality, in columns 1-2 of Table A.1, we replicate the results in Table 2 and further control the lagged stock returns. In columns 1-2 of Table A.2, we include more risk factors, such as Robust Minus Weak, Conservative Minus Aggressive and momentum. The results are similar as those in Table 2: Posting a Tweet that contains 10% of positive words, the daily return will increase by 1.3 basis points.

[Insert Table 3 about here]

Electronic copy available at: https://ssrn.com/abstract=3293134

Next, we include stock returns on days without CEOs posting Tweets. On days without Tweets, the proportion of positive or negative words is zero and the total number of words is also zero. Table 3 reports the regression results adding the days without Tweets during CEO tenure. We include the same controls and fixed effects as in Table 2. Panel A reports the regression results which drop Tweets related to firm operations, while Panel B keeps all of the Tweets. The coefficients of the proportion of positive words are around 0.0012 to 0.0018 with various fixed effects and they are significant at the 1% level of significance. The interpretation is that posting a Tweet with an average proportion (10%) of positive words will increase stock returns by between 1.3 and 1.9 basis points, as compared to the days that CEOs do not post Tweets, or the days that CEOs post Tweets that do not contain positive words. These results show that the proportion of positive words in Tweets can predict stock returns. Moreover, columns 3-4 in Table A.1 and A.2 replicate the regression results in Table 3 by controlling the lagged stock performance and risk factors, respectively. The magnitudes of the coefficients in Table A.1 and A.2 are similar to the results in Table 3.

[Insert Table 4 about here]

To further explore the predictability of Tweets on later stock returns, we regress the proportion of positive words on daily stock returns subsequent to the Tweet days. Specifically, the dependent variables in Table 4 include stock returns on the first, second, third and sixth day after CEOs posting the Tweets. We include the same control variables as well as fixed effects as in the last column of Table 2. Panel A excludes Tweets that are related to firm operations and Panel B contains all the Tweets. The first three columns show that there is predictive power of positive words on subsequent stock returns. To be specific, the coefficients of the proportion of positive words are 0.11%, 0.09% and 0.10% on the first, second and third day following the Tweets. However, the increasing trend of stock returns reverses on the sixth day after the Tweets days, as the coefficient of the proportion of positive words is negative on the sixth day's return. Therefore, we find that

Electronic copy available at: https://ssrn.com/abstract=3293134

the proportion of positive words can predict contemporary stock returns using Tweets posted on general days.

3.2 Stock Returns Before Earnings Announcements

In this section, we explore the predictive power of positive words in CEO Tweets prior to earnings announcements. If the proportion of positive words in the Tweets of CEOs is a proxy for positive information about firm performance, we expect this predictive power for stock returns should be stronger before earnings announcements. This is because CEOs already know whether their firms' earnings have beaten analysts' forecasts, while investors are not informed of such information. Compared to CEOs who have failed to meet the analysts forecasts, CEOs who have achieved the performance target are more likely to be happier and engage in enjoyable activities. Therefore, we can use CEO Tweets to infer the mood of CEOs before earnings release.

[Insert Table 5 about here]

To start with, we predict contemporary stock returns before earnings announcements. Table 5 presents the regression results of CEOs' Tweets on stock returns during two weeks before earnings announcements. We include the same control variables and fixed effects as in Table 2. Panel A reports the regression results by excluding Tweets related to firm operations, while Panel B includes all of the Tweets. The coefficients of the proportion of positive words are around 0.0045 before earnings announcements, which are triple the positive word coefficients in Table 3 that covers the whole sample period. The predictive power is magnified before earnings release due to the reason that investors are waiting for news about actual earnings, thus these stakeholders are more sensitive and pay a greater amount of attention to communications and information released from CEOs. Moreover, we further include the lagged stock returns in columns 5-6 of Table A.1. and more risk factors, such as Robust Minus Weak, Conservative Minus Aggressive and momentum in Columns 5-6 of Table A.2. The coefficients of the proportion of positive words are also

Electronic copy available at: https://ssrn.com/abstract=3293134

around 0.4% on stock returns before earnings announcements.

[Insert Figure 2 about here]

Furthermore, using an event study framework, we explore the cumulative abnormal returns (CAR) around Tweet days before earnings announcements. The abnormal return is measured as the difference of daily stock return and the equal-weighted market return. The cumulative abnormal return is the sum of abnormal returns starting 14 days before CEOs post the Tweets to 14 days afterwards. Figure 2 plots the CAR separately for three groups. The first group contains days with no Tweets posted and they appear as red squares. The second group includes days with Tweets posted that have no positive words and they appear as blue rhombuses. The third group is consisted of days with Tweets that contain positive words and they appear as green triangles. The group of days with Tweets containing positive words shows an increasing trend of stock returns around Tweet days compared to the other two groups. Moreover, this increasing pattern persists without a reversal during a 28-day window. Therefore, this event study provides some evidence that there is information content in the Tweets of CEOs posted before earnings announcements. Specifically, by tracking CEOs' Tweets, we can infer CEOs' activities and consequently whether they have achieved analysts forecasts. Executives who perform well tend to use more positive words in their Tweets, which can predict higher contemporary returns on the Tweet days and subsequent stock returns.

[Insert Table 6 about here]

In addition to predicting stock returns on Tweet days, we predict stock returns on earnings announcements using positive words in Tweets posted prior to earnings announcements. Table 6 presents the predictive power of positive words in Tweets on subsequent stock returns. Specifically, the dependent variable is the daily stock return on earnings release days, while the independent variable is the proportion of positive words in Tweets posted during 14 days prior to the announcement. Panel A includes Tweets that are

Electronic copy available at: https://ssrn.com/abstract=3293134

unrelated to firm operations and Panel B contains all the Tweets. We control for the same variables and fixed effect as in the first two columns of Table 2. The coefficients of the proportion of positive words are 0.7% to 0.9% on stock returns on earnings release, and they are significant at the 5% level. It indicates that posting a Tweet containing a 7% fraction of positive words predicts 5 basis points higher in stock returns on earnings announcement days.

[Insert Table 7 about here]

Alternatively, we can aggregate the proportion of positive words in Tweets posted prior to earnings announcements to predict stock returns on those days. Table 7 shows the regression results. The independent variable of interest is the cumulative proportion of positive words, which is counted starting 14 days before, and ending 1 day prior to, earnings announcements. We control for the cumulative proportion of negative words, cumulative number of Tweets, Fama-French factors, firm and CEO features, as well as industry and time fixed effects. Panel A excludes Tweets related to firm operations in the calculation of cumulative measurements, while Panel B contains all of the Tweets. The coefficients of the cumulative proportion of positive words are positive on stock returns on earnings announcement days at the significance level of 10%.

In summary, the predictive power of CEO Tweets for contemporary and subsequent returns is stronger before earnings announcements, compared to the predictive power of Tweets for stock returns on regular days. When CEOs are confident and satisfied with firm operations, they tend to engage in pleasant activities and use more positive words to tweet. Therefore, we can use this information to predict stock returns on the Tweet days prior to earnings release as well as returns on earnings announcement days.

3.3 Portfolio Returns

In the previous section, we show that there exists information in CEOs' Tweets prior to earnings announcements based on two findings. First, the increasing trend of stock returns on Tweet days persists without a reversal in a window of 28 days. Second, we can use the

Electronic copy available at: https://ssrn.com/abstract=3293134

positive words in Tweets posted during the 14 days prior to earnings release to predict stock returns on earnings announcement days. Thus, positive words have predictive power, not only for contemporary stock returns, but also for subsequent returns on earnings release days. In this section, we further construct investment portfolios based on Tweets posted prior to earnings announcements and examine the performance of these portfolios.

Because stock returns increase as the proportion of positive words increases, we long portfolios of stocks that contain positive words in their CEO Tweets, and short stocks that lack positive words. Specifically, we restrict the sample to be Tweets posted during the 14 days prior to earnings release, and sort the Tweets based on the proportion of positive words on Tweet days. If there are multiple Tweets that have an equal fraction of positive words, we further sort them based on the proportion of negative words. We further restrict to days that contain at least two Tweets, including one Tweet with positive words.

[Insert Table 8 about here]

Table 8 reports Fama-French factor alphas on daily calendar-time spread portfolios that buy stocks that contain a high proportion of positive words in their CEOs' Tweets and sell stocks that contain a low fraction of positive words during the 14 days prior to earnings announcements. The sample contains 988 days that have at least two Tweets posted, including at least one Tweet with positive words. We sort the sample based on the proportion of positive words on Tweet days and rebalance the portfolio on a daily basis. We track the performance of these portfolios over the next day, as well as over the following week. We apply two weighting schemes: equal weighting and value weighting.

First, we use an equal weighting scheme and buy stocks with the proportion of positive words in the top 10th percentile and sell stocks in the bottom 10th percentile. Panel A describes the summary statistics of these stocks. For stocks that fall in the top and bottom 10th percentile of the proportion of positive words, their portfolio returns over the next day are 0.21% and 0.056%, respectively. Each portfolio contains 1.24 stocks on average. Panel B reports alphas and factor loadings by regressing daily stock returns on risk factors

Electronic copy available at: https://ssrn.com/abstract=3293134

using an equal weighting scheme. A portfolio of stocks with the proportion of positive words in the top 10th percentile has an alpha of 12.8 basis points per day. In contrast, the alpha for the portfolio of stocks in the bottom 10th percentile is -4.4 basis points. Therefore, a spread portfolio that buys stocks in the top 10th percentile of the proportion of positive words and sells stocks in the bottom 10th percentile has an alpha of 17.2 basis points per day, which is significantly positive at 1% level.

Furthermore, we apply a value-weighted scheme in calculating the portfolio returns. Panel C replicates the investment strategy in Panel B using a value-weighting scheme. The portfolio of stocks with the proportion of positive words in the top or bottom 10th percentile has an alpha of 13.3 and -6.3 basis points, respectively. A daily calendar-time spread portfolio based on a value-weighted scheme that longs stocks in the top 10th percentile of positive words and shorts stocks in the bottom 10th percentile has an alpha of 19.6 basis points.

We also track the performance of these portfolios over the following week. To be specific, in Panel D, we report the alpha and factor loadings of the calendar-time portfolios in Panel B with a holding period of a week. The portfolios are rebalanced daily by adding stocks with positive words that enter in the top or bottom percentiles and dropping stocks that reach the end of the holding period. Stocks in the top 10th percentile have an alpha of 12.5 basis points, while stocks that fall in the bottom 10th percentile have an alpha of 1.6 basis points. Therefore, a spread portfolio that longs stocks in the top 10th percentile of the proportion of positive words and shorts stocks in the bottom 10th percentile with a holding period of a week has an alpha of 10.8 basis points per day.

4 Using CEOs' Tweets to Predict Earnings

4.1 Predicting Earnings Surprise

In this section, we investigate whether the Tweets of CEOs can predict firms' earnings surprise. On one hand, if the fraction of positive words in Tweets is a noise measurement of positive information about firms, the coefficients of positive words in the regressions

Electronic copy available at: https://ssrn.com/abstract=3293134

on firms' earnings surprise should be around zero. On the other hand, if CEOs tend to use more positive words in their Tweets when they are confident and satisfied with firm performance, the coefficients on earnings surprise should be significantly positive. Thus, positive words such as "gorgeous, incredible, good, happy, nice, terrific, amazing, and wonderful" should predict contemporaneous firm earnings.

We examine Tweets before earnings announcements to predict the difference between actual earnings and analysts' forecasts. We choose a window of 14 days before earnings announcements, given that CEOs already know whether their firms will beat analysts' forecasts, even though this information has not been released to public at that time. With regard to firms that meet analysts' forecasts, CEOs may spend more time with their families, watching sports, hanging out with friends, travelling on vacations or they may simply be happier than CEOs of firms that miss their earnings target. In terms of Tweets, enjoyable activities and a pleasant mood are reflected as positive words in their postings.

[Insert Table 9 about here]

Table 9 presents the relationship between earnings surprise and cumulative measurements of Tweets before earnings announcements. These cumulative measurements about Tweets are counted starting 14 days before, and ending 1 day prior to, earnings announcements. The independent variable of interest is the cumulative proportion of positive words. We also control the cumulative proportion of negative words and the total number of Tweets posted during this period. The dependent variable is the standardized unexpected earnings, which is collected in the surprise history of the summary history of the Institutional Brokers' Estimate System (IBES). Standardized unexpected earnings are defined as the difference of actual quarterly earnings and the mean of analysts' forecasts, which is known as "earnings surprise," divided by the standard deviation of the earnings surprise. The summary statistics of the cumulative measurements of Tweets, and earnings surprise can be found in Panel C of Table 1. For example, the mean of the standardized

Electronic copy available at: https://ssrn.com/abstract=3293134

unexpected earnings is 1.633. The cumulative proportion of positive words for the 14 days before earnings announcements is 0.02.

With these definitions, the regression equation can be expressed as:

$$\begin{aligned}\text{Standardized Unexpected Earnings } &= \alpha + \beta_1 \text{Cum. Pos. Prop.} + \beta_2 \text{Cum. Neg. Prop.} \\ &\quad + \beta_3 \text{Cum. Num. Tweets} + \text{Controls}\end{aligned}$$

Panel A in Table 9 excludes Tweets related to firm operations. We include industry (firm) and year fixed effects in columns 1 (3), and industry (firm), year and quarter fixed effects in Column 2 (4). The coefficients of the cumulative proportion of positive words are around 0.78 to 0.94, and they are significant at the 10% level. If the cumulative proportion of positive words increased by 1%, the standardized unexpected earnings would increase by 0.01. Panel B of Table 9 includes all of the Tweets posted during the 14 days before earnings announcements. The coefficients of the cumulative proportion of positive words are slightly larger than those in Panel A, and they remain significant at the 10% of level of significance. Thus, these findings show that the cumulative proportion of positive words in CEOs' Tweets prior to earnings announcements can predict earnings surprise.

[Insert Table 10 about here]

Next, we examine the cumulative number of positive words in CEOs' Tweets posted in the two weeks preceding the earnings release. Table 10 employs the following regression:

$$\begin{aligned}\text{Standardized Unexpected Earnings } &= \alpha + \beta_1 \text{Cum. Pos. Num.} + \beta_2 \text{Cum. Neg. Num.} \\ &\quad + \beta_3 \text{Cum. Num. Tweets} + \text{Controls}\end{aligned}$$

The independent variable of interest is the cumulative number of positive words. We control the cumulative number of negative words and the cumulative number of Tweets. We include the same controls on firm and CEO characteristics and fixed effects as in Table 9. Panel A reports the regression results from dropping Tweets related to firm operations,

Electronic copy available at: https://ssrn.com/abstract=3293134

while Panel B keeps all of the Tweets. The coefficients of the cumulative number of positive words are around 0.11 and they are significant at the 5% level of significance. This means that adding one positive word will increase the standardized expected earnings by 0.11, which is equivalent to 10% of the median of standardized unexpected earnings (1.154). Thus, the cumulative number of positive words preceding earnings announcements can predict earnings surprises.

[Insert Table 11 about here]

The previous analysis examines CEOs' Tweets during a window of 14 days preceding the earnings announcements. Alternatively, Table 11 presents the results of predicting the earnings surprise using Tweets within 28 days, or 7 days prior to the earnings release. In Table 11, the independent variable of interest in Panel A is the cumulative proportion of positive words. Column 1 and 2 include Tweets within one week prior to earnings announcements, and columns 3 and 4 contain Tweets during a window of 28 days preceding the earnings release. The coefficients of the cumulative proportion of positive words are 1.04 and 0.64 for Tweets during a window of 7 and 28 days, respectively. Recall that the coefficient of the cumulative proportion of positive words in a window of 14 days is 0.93 in Panel B of Table 9. These findings show that the information content of CEOs' Tweets is stronger when the Tweets posted on days close to earnings announcements. In Panel B of Table 11, we use the cumulative number of positive words to predict the earnings surprise. In columns 1 and 2, the coefficients of the cumulative number of positive words are 0.11 during a 7-day timeframe and 0.07 during a 28-day window. The coefficients of the cumulative number of positive words on earnings surprise in a 14-day framework are also 0.11, according to Panel B of Table 10. In summary, the above findings are consistent with information content theory concerning CEOs' Tweets, which can be used to predict earnings surprise.

[Insert Table 12 about here]

Electronic copy available at: https://ssrn.com/abstract=3293134

4.2 Predicting Firms' Fundamentals

In this section, we investigate whether the Tweets of CEOs can predict firms' fundamentals such as earnings per share, normalized earnings and cash flows.We first investigate the relationship between quarterly earnings per share and CEOs' Tweets preceding the end of each quarter In Table 12. Earnings per share are measured as the diluted EPS dividend by lagged stock price, basic EPS dividend by lagged stock price, income before extraordinary items divided by lagged capitalization, income before extraordinary items adjusted for common stock equivalents divided by lagged capitalization, or income before extraordinary items available for common divided by lagged capitalization. The independent variable of interest is the cumulative proportion of positive words in Tweets during the 14 days prior to the end of each quarter. Panel D of Table 1 presents the summary statistics of these outcome variables on quarterly earnings per share, together with information about Tweets posted during the two weeks prior to the end of each quarter. For example, the mean of the cumulative proportion of positive words is 0.025 and the average number of Tweets posted is 3.77.

The regression equation specification in Table 12 is as the following:

$$\begin{aligned}\text{Normalized EPS} = \alpha + \beta_1 &\text{Cum. Pos. Prop} + \beta_2 \text{Cum. Neg. Prop.} \\ &+ \beta_3 \text{Cum. Num. Tweets} + \text{Controls}\end{aligned}$$

We include firm and year fixed effects in the odd columns and firm, year and quarter fixed effects in the even columns. Panel A excludes Tweets about firm activities, while Panel B contains all of the Tweets. The coefficients of the cumulative proportion of Tweets are around 0.034, and they are significant at the 5% level. This means that a 1% increase in the cumulative proportion of positive Tweets will increase the normalized earnings per share by 0.0003, which is equal to 10% of the mean of normalized earnings per share (0.004). In other words, having a higher proportion of positive words in the Tweets of CEOs prior to the end of each quarter is associated with a higher level of earnings per

Electronic copy available at: https://ssrn.com/abstract=3293134

share.

[Insert Table 13 about here]

Last, we explore the predictability of CEOs' Tweets with respect to normalized earnings. Table 13 regresses cumulative measurements of CEOs' Tweets within 14 days prior to the end of each quarter on normalized earnings. Normalized earnings are defined as 4 times net income divided by lagged assets, income before extraordinary items divided by lagged assets, income before extraordinary items adjusted for common stock equivalents divided by lagged assets, income before extraordinary items available for common divided by lagged assets, or cash flow divided by lagged assets. The control variables and fixed effects are the same as in Table 12. Panel A excludes Tweets about firm operations, and Panel B contains all of the Tweets. The coefficient of the cumulative proportion of positive words is 0.09 for normalized earnings. Given that the mean of net income divided by assets is 0.04, a 1% increase in the cumulative proportion of positive words would enhance the normalized earnings by 0.001, or 3% of the mean. The coefficients on cash flow are 0.022 in columns 9 and 10, meaning that a 1% increase in the cumulative proportion of positive words would increase the normalized cash flow by 1% of its mean.

In summary, the above findings show that the Tweets of CEOs contain information about the firms' fundamentals. The cumulative measurements of positive words can predict earnings surprise, earnings per share and normalized earnings. When CEOs are confident and satisfied with firm performance, they are happier and spend more time on personal activities. Their pleasant mood and enjoyable activities are reflected in Tweets using positive words, such as "gorgeous, incredible, good, happy, nice, terrific, amazing, and wonderful."

5 Motivation Behind The Tweets

So far, we have shown the information content of the Tweets of CEOs, which can predict a higher level of stock returns and earnings. In this section, we explore whether CEOs

Electronic copy available at: https://ssrn.com/abstract=3293134

Tweets may constitute unintentional release of information without an agenda, or it may be that CEOs strategically choose the time to release the information.

Literature has found that managers manipulate the timing of corporate voluntary disclosures as a mean of increasing the fair value of executive compensation. These studies focus on lowering stock prices on option grant dates, given that option exercise prices are set to be stock prices on award dates. In a manner consistent with this reasoning, Aboody and Kasznik (2000) find that CEOs manage to rush forward bad news and delay announcements of good news on earnings prior to option award dates. Chauvin and Shenoy (2001) also show that stock prices prior to option grants tend to decrease due to the release of bad discretionary news on earnings, dividends and forecasts. Yermack (1997) examines stock price movements after award dates and finds that prices increase following option grants due to favorable earnings announcements.

This paper examines another communication channel through which executives transmit information to investors, which is the personal social media of CEOs. According to Chauvin and Shenoy (2001), executives do not want to spoil their compensation packages before the compensation committee meetings by formally releasing bad news. Thus, executives manipulate stock prices using informal or private communication methods, given that the resulting stock prices can be blamed on market misinformation. The personal social media of CEOs is an informal communication tool, and we have shown that there is information content in CEOs' Tweets. In particular, having a higher proportion of positive words is associated with higher stock returns. Therefore, if CEOs strategically time the disclosure of information to increase their personal payoff, CEOs would post Tweets that contain a higher fraction of positive words when they want to increase stock prices. Thus, we compare the pattern of Tweets that are posted around CEO option exercises and stock sales.

Dates of CEO stock option exercises and stock sales are collected from Table 2 and Table 1 of the Thomson Reuters database. We use managers' role titles to restrict managers to CEOs only. We limit the Tweets posted during the 14 days before and the 14 days after the dates of option exercises and stock sales. Panels E and F in Table 1 present the summary statistics for the proportion of positive and negative words, and the number of

Electronic copy available at: https://ssrn.com/abstract=3293134

words in each Tweet posted before and after option exercises and stock sales, respectively. For example, the average proportion of positive words is 0.017 before option exercises and it is 0.013 after option exercises, and the difference is significant at the 1% level. The mean of proportion of negative words is also significantly higher before than after option exercises, but it becomes insignificant once we control for the firm fixed effect in a regression framework.

[Insert Table 14 about here]

Panel A of Table 14 explores the pattern of Tweets posted during the 14 days before and the 14 days after CEO option exercises using the following regression equation:

$$\text{Characteristics of Tweets} = \alpha + \beta_1 \text{I(Before Option Exercise)} + \text{Controls}$$

Column 1 and 2 examine the proportion of positive words, columns 3 and 4 explore the proportion of negative words, and columns 5 and 6 investigate the total number of words in each Tweet. The independent variable of interest is the before option exercise indicator, which takes on a value of 1 if a Tweet is posted within 14 days prior to option exercises, and it is equal to 0 if a Tweet occurs within 14 days after option exercises. The control variables and fixed effects are the same as in Table 13. The coefficient of the before option exercise indicator on the proportion of positive words is 0.0012, and it is significant at the 10% level. It means that CEOs posted a 1% higher fraction of positive words before option exercises than afterwards. The coefficient of before exercise indicator is insignificant with respect to the proportion of negative words. Thus, it provides some evidence that CEOs do not increase the fraction of positive and negative words simultaneously before exercises. The last two columns show that the number of words in Tweets is 0.33 more before than after option exercises.

Similarly, Panel B of Table 14 presents the pattern of CEOs' Tweets around the time of stock sales. The independent variable of interest is the before stock sale indicator, which takes on a value of 1 if a Tweet is posted during the 14 days prior to stock sales, and it

Electronic copy available at: https://ssrn.com/abstract=3293134

is equal to 0 if a Tweet occurs within 14 days after stock sales. The coefficient of before stock sale indicator on the proportion of positive words is 0.0013 and it is significant at the 5% level. The coefficient on the proportion of negative words is 0.002 and it is not significant at the 10% level. These results show that CEOs post Tweets that contain a higher proportion of positive words, but not a higher proportion of negative words, before they sell their stocks.

The above findings provide some evidence that CEOs strategically post Tweets that contain a higher proportion of positive words in order to manipulate stock price upward before option exercises and stock sales. These results are consistent with the literature showing that executives are incentivized to inflate stock prices leading up to option exercises. For example, Balachandran et al. (2008) suggest that firms use buyback programs to increase stock prices and maximize payoff from option exercises.

6 Concluding Remarks

Using one hundred thousand personal Tweets of CEOs posted during their tenure, we test four competing hypotheses about the relationship of CEOs' Tweets and market reaction. First, CEOs' personal Tweets are irrelevant with respect to firms' fundamentals, or the information about firms' fundamentals is already incorporated into stock prices. If the fraction of positive words in Tweets is a noise measurement of positive information about firms, the coefficient of positive words in the regressions on firms' fundamentals should be zero. Similarly, if the proportion of positive words in Tweets is a proxy for positive information about firms' fundamentals that is already incorporated into prices, we should find no predictive power for future market activities. Given that we document a significant positive relationship between the proportion of positive words in Tweets and firms' stock returns and earnings, we reject this hypothesis.

Second, sentiment theory holds that irrational investors misinterpret the positive words in Tweets and buy stocks, causing a stationary increase in returns. If the higher stock returns are due to sentiment, we should see returns rebound during the next period when there is a new information about firms. However, we do not observe the CAR around earn-

Electronic copy available at: https://ssrn.com/abstract=3293134

ings announcements reverses during a 28-day window. In addition, based on sentiment theory, we should not find there to be predictability concerning CEOs' Tweets regarding firms' fundamentals, such as earnings surprises. Third, the conspiracy hypothesis contends that a temporary increase in stock price is caused by false information due to CEO corruption. If higher stock returns are the results of conspiracy, we should find that stock prices reverse when investors discover the true information. Given that we find that CAR around earnings announcements increases persistently, and that Tweets can predict firms' earnings, we reject both the sentiment and conspiracy hypotheses.

Fourth, information theory believes that positive words in Tweets are proxies about firms' fundamentals that are not currently incorporated into stock prices. My findings support the information content hypothesis. We find that the proportion of positive words in CEOs' Tweets can predict stock returns, especially before earnings announcements. Furthermore, the cumulative measurements of positive words can predict earnings surprise, earnings per share and normalized earnings. Intuitively, when CEOs are confident and satisfied with their firms' performance, they tend to tweet using positive tones about their daily lives. Messages containing positive words derive from pleasant mood or enjoyable activities, such as dining out, travelling, watching sports, spending time with family, hanging out with friends and appreciating the scenery.

This paper is closed related to Yermack (2014). Tracking corporate jet flights, he finds a connection between CEO vacations and the timing of corporate information release. He shows that corporations withhold good news and release it immediately after CEOs return from vacations. Compared to his measurement of CEOs' private activities using jet flights, my measurement using the Tweets of CEOs is simpler to measure and has a higher frequency. To be specific, flights may happen infrequently, whereas Tweets of CEOs could occur daily. Counting the proportion of positive words in Tweets can allow investors to infer CEOs' job achievements and attainability, such as whether CEOs have achieved the earnings target goal. These findings can be used to develop trading strategies. For example, using Tweets posted before earnings announcements, a daily calendar-time spread portfolio that buys stocks with a high proportion of positive words and sells stocks with a low proportion delivers a return of 10.8 to 19.6 basis points per day. However, the

Electronic copy available at: https://ssrn.com/abstract=3293134

limitation of this paper is the sample selection bias. Only 10% of CEOs have their own personal Twitter accounts.

In comparing the literature on information content, this study relies on the personal social media of CEOs, while the literature focuses on media coverage, analysts' forecasts and firms' publicly disclosed accounting variables. We measure the tone of CEOs' Tweets by tracking positive words in order to infer CEO job performance and thus business prospects. This is different than the traditional data sources that directly report firms' activities and operations.

Electronic copy available at: https://ssrn.com/abstract=3293134

References

Aboody, D., and R. Kasznik. 2000. CEO stock option awards and the timing of corporate voluntary disclosures. *Journal of accounting and economics* 29:73–100.

Antweiler, W., and M. Frank. 2006. Do US stock markets typically overreact to corporate news stories? .

Antweiler, W., and M. Z. Frank. 2004. Is all that talk just noise? The information content of internet stock message boards. *The Journal of finance* 59:1259–1294.

Baker, T., D. Collins, and A. Reitenga. 2003. Stock option compensation and earnings management incentives. *Journal of Accounting, Auditing & Finance* 18:557–582.

Balachandran, B., K. Chalmers, and J. Haman. 2008. On-market share buybacks, exercisable share options and earnings management. *Accounting & Finance* 48:25–49.

Bartov, E., and P. Mohanram. 2004. Private information, earnings manipulations, and executive stock-option exercises. *The Accounting Review* 79:889–920.

Chauvin, K. W., and C. Shenoy. 2001. Stock price decreases prior to executive stock option grants. *Journal of Corporate Finance* 7:53–76.

Davis, A. K., J. M. Piger, L. M. Sedor, et al. 2006. Beyond the numbers: An analysis of optimistic and pessimistic language in earnings press releases. Tech. rep., Federal Reserve Bank of St. Louis.

Erel, I., L. H. Stern, C. Tan, and M. S. Weisbach. 2018. Selecting Directors Using Machine Learning. Tech. rep., National Bureau of Economic Research.

Huang, J. 2018. The customer knows best: The investment value of consumer opinions. *Journal of Financial Economics* 128:164–182.

Li, F. 2006. Do stock market investors understand the risk sentiment of corporate annual reports? .

Electronic copy available at: https://ssrn.com/abstract=3293134

Luca, M. 2011. Reviews, reputation, and revenue: The case of Yelp. com. Harvard Business School NOM Unit Working Paper 12-016, 1-40.

Tetlock, P. C. 2007. Giving content to investor sentiment: The role of media in the stock market. *The Journal of finance* 62:1139–1168.

Tetlock, P. C., M. Saar-Tsechansky, and S. Macskassy. 2008. More than words: Quantifying language to measure firms' fundamentals. *The Journal of Finance* 63:1437–1467.

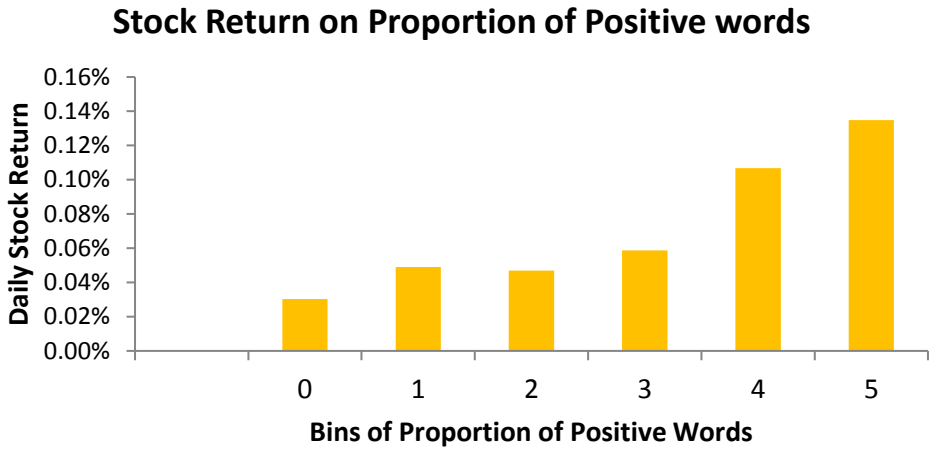
Yermack, D. 1997. Good timing: CEO stock option awards and company news announcements. *The journal of Finance* 52:449–476.

Yermack, D. 2014. Tailspotting: Identifying and profiting from CEO vacation trips. *Journal of Financial Economics* 113:252–269.

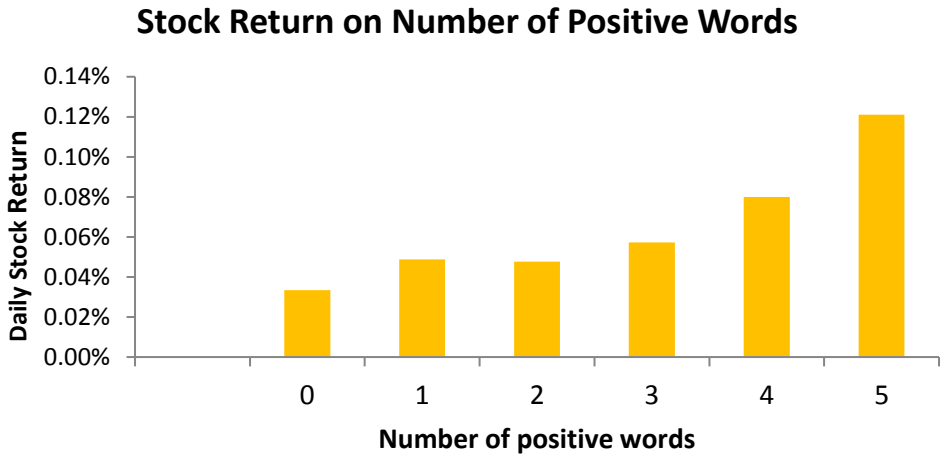
Electronic copy available at: https://ssrn.com/abstract=3293134

Figure 1 Daily stock returns and positive words in Tweets
The following graphs plot the relationship between positive words in Tweets and stock returns. Panel A plots the daily stock returns after CEOs posting Tweets on bins of the proportion of positive words in Tweets. The proportion of positive words is defined as the number of positive words divided by the total number of words in each Tweet. Bin 0 includes Tweets with zero positive word, bin 1 contains Tweets with (0, 0.1] proportion of positive words, bin2 to bin 4 have (0.1, 0.2], (0.2, 0.3] and (0.3, 0.4] proportion, respectively, and bin 5 contains Tweets with more than 0.4 fraction. Panel B plots the daily stock returns on the number of positive words. If a Tweet contains more than 5 positive words, it is included in the last bar with 5 positive words.

A: Proportion of positive words

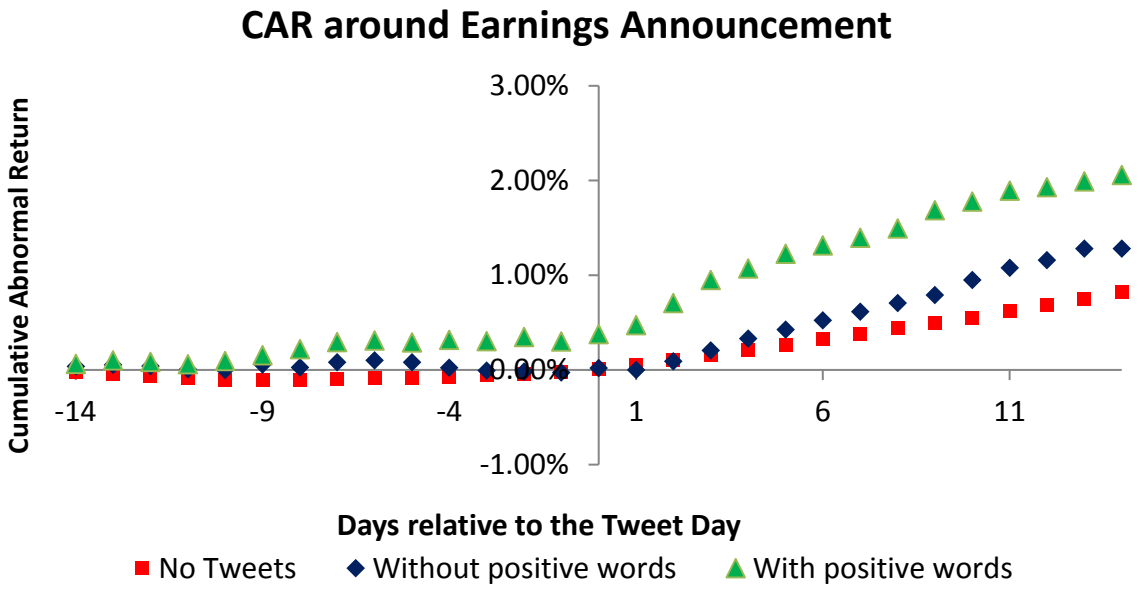


B: Number of positive words



Electronic copy available at: https://ssrn.com/abstract=3293134

Figure 2 Cumulative abnormal returns around Tweet days before earnings announcements
The following graphs plot the cumulative abnormal return around the Tweet days before earnings announcements. The abnormal return is measured as the difference between the daily stock return and the equal-weighted market return. The cumulative abnormal return is the sum of abnormal returns 14 days before CEOs post the Tweets to 14 days afterwards. Cumulative abnormal returns are plotted separately for three groups: days with no Tweets posted (red squares), days with Tweets posted that contain no positive word (blue rhombus), and days with Tweets posted that contain positive words (green triangles).



Electronic copy available at: https://ssrn.com/abstract=3293134

Table 1 Summary statistics

The following tables present summary statistics. Panel A describes stock return, Fama-French 3 factors, and Tweet information on days that CEOs post Tweets. Panel B reports firm and CEO characteristics, which are measured at the end of the previous year. Panel C reports summary statistics on earnings announcement days, which includes the cumulative measurements of Tweets. The cumulative measurements are counted starting 14 days to 1 day prior to earnings announcements. Panel D reports other outcome variables on firms' fundamentals, such as earnings per share, normalized earnings and cash flow. These outcome variables are measured at the end of each quarter. The cumulative measurements of Tweets are counted 14 days to 1 day prior to the end of each quarter. Panel E compares information on Tweets that are posted during 14 days before CEO option exercises with those posted during 14 days after option exercises. Panel F reported the summary statistics of Tweets that are posted during 14 days prior to CEO stock sales and during 14 days after stock sales.

A: On Tweet days (N=127,916)

	P10	P25	P50	P75	P90	Mean	Std
Daily stock return (%)	-2.234	-0.914	0.023	1.063	2.356	0.075	0.021
Small minus big	-0.600	-0.310	0.010	0.330	0.630	0.012	0.517
High minus low	-0.570	-0.310	-0.040	0.260	0.540	-0.017	0.486
Market premium	-1.030	-0.330	0.060	0.500	1.100	0.055	0.938
Pos. prop	0.000	0.000	0.053	0.111	0.174	0.072	0.102
Neg. prop	0.000	0.000	0.000	0.000	0.071	0.018	0.049
Total words	5.000	9.000	14.000	19.000	23.000	14.191	7.414
I(firms' operation related)	0.000	0.000	0.000	0.000	0.000	0.013	0.113

B: Firm and CEO characteristics (N=819)

	P10	P25	P50	P75	P90	Mean	Std
Cash to assets	0.020	0.061	0.137	0.291	0.468	0.200	0.188
Total assets ($million)	233	592	2797	13020	65444	19950	42074
Market to book	1.188	1.916	3.129	5.599	10.878	5.197	7.029
Return on assets	-0.042	0.010	0.038	0.077	0.132	0.037	0.101
Market leverage	0.082	0.156	0.287	0.487	0.694	0.342	0.229
CEO total comp.($thousand)	1137	2292	4880	10155	17569	6832	5737
CEO age	44.000	48.000	52.000	57.000	61.000	52.631	7.132
CEO tenure	0.000	1.000	5.000	11.000	17.000	7.196	7.728

C: On earnings announcement days (N=4,544)

	P10	P25	P50	P75	P90	Mean	Std
Stand. unexpected Earnings	-1.761	-0.167	1.154	2.959	5.495	1.633	7.813
Cum. pos. num.	0.000	0.000	0.000	1.000	5.000	1.309	2.917
Cum. neg. num	0.000	0.000	0.000	0.000	1.000	0.431	1.533
Cum. Tweets	0.000	0.000	0.000	2.000	7.000	1.858	4.004
Cum. words	0.000	0.000	0.000	24.000	97.000	26.121	56.403
Cum. pos. prop.	0.000	0.000	0.000	0.025	0.074	0.020	0.042
Cum. neg. prop.	0.000	0.000	0.000	0.000	0.018	0.005	0.016

Electronic copy available at: https://ssrn.com/abstract=3293134

Cont. of Table 1

D: Other fundamental outcome variables (N=5,306)

	P10	P25	P50	P75	P90	Mean	Std
Diluted EPS / price	0.011	0.003	0.009	0.016	0.024	0.004	0.044
Basic EPS / price	-0.011	0.003	0.009	0.016	0.024	0.004	0.044
Income before /cap.	-0.011	0.003	0.010	0.016	0.024	0.004	0.044
Income before adj. / cap.	-0.011	0.003	0.009	0.016	0.024	0.004	0.044
Income before for common / cap.	-0.011	0.003	0.009	0.016	0.024	0.004	0.044
net income/assets	-0.050	0.009	0.043	0.087	0.149	0.038	0.148
Income before /assets	-0.046	0.009	0.043	0.085	0.148	0.039	0.142
Income before adj. / assets	-0.047	0.009	0.043	0.085	0.148	0.038	0.143
Income before for common / assets	-0.047	0.009	0.043	0.085	0.148	0.038	0.143
Cash flow / assets	-0.002	0.010	0.021	0.033	0.050	0.020	0.037
Cum. pos. num.	0.000	0.000	0.000	2.000	10.000	3.532	11.971
Cum. neg. num	0.000	0.000	0.000	0.000	2.000	0.947	5.325
Cum. Tweets	0.000	0.000	0.000	2.000	9.000	3.769	14.263
Cum. words	0.000	0.000	0.000	27.000	142.000	54.186	198.810
Cum. pos. prop.	0.000	0.000	0.000	0.042	0.088	0.025	0.047
Cum. neg. prop.	0.000	0.000	0.000	0.000	0.016	0.005	0.016

E: Tweets around option exercises

	Before and after option exercises		Before option exercise		After option exercise		Difference
	P50	Mean	P50	Mean	P50	Mean	
Pos. prop	0.000	0.016	0.000	0.017	0.000	0.013	0.005***
Neg. prop	0.000	0.003	0.000	0.004	0.000	0.003	0.001***
Total words	0.000	3.195	0.000	3.521	0.000	2.677	0.843***
N	30,933	30,933	18,984	18,984	11,949	11,949	

F: Tweets around stock sales

	Before and after stock sale		Before stock sale		After stock sale		Difference
	P50	Mean	P50	Mean	P50	Mean	
Pos. prop	0.000	0.019	0.000	0.021	0.000	0.015	0.006***
Neg. prop	0.000	0.005	0.000	0.005	0.000	0.004	0.001***
Total words	0.000	3.781	0.000	4.101	0.000	3.225	0.877***
N	55,331	55,331	35,118	35,118	20,213	20,213	

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 2 Predicting stock returns on days with Tweets

The following tables report the relation between daily stock returns and the fraction of positive words in Tweets. The proportion of positive words is defined as the number of positive words divided by the total number of words in a Tweet. The controls include the proportion of negative words, the total number of words, Fama-French factors, firm characteristics, and CEO characteristics. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We include industry or firm interacting with time fixed effects. Error is clustered at the firm and year level. T-statistics are reported in the parenthesis.

A: Exclude firm operations related Tweets

	Stock Return on Tweet day (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.131**	0.130**	0.106**	0.124**
	(2.58)	(2.35)	(2.40)	(2.79)
Neg. prop.	-0.072	-0.082	-0.14	-0.193*
	(-0.57)	(-0.67)	(-1.45)	(-2.12)
Total words	-0.000	-0.001	0.001	0.001
	(-0.23)	(-0.37)	(0.62)	(0.74)
Small minus big	0.339***	0.324***	0.336***	0.325***
	(4.32)	(4.28)	(4.48)	(4.44)
High minus low	-0.008	-0.002	-0.023	-0.015
	(-0.14)	(-0.03)	(-0.41)	(-0.26)
Market risk	0.764***	0.764***	0.761***	0.756***
premium	(19.40)	(20.69)	(20.67)	(21.04)
R square	0.171	0.205	0.186	0.246
N	117,332	117,233	124,827	124,251
Control	Yes	Yes	No	No
Fixed effect	Ind*Y*Q	Ind*Y*M	Firm*Y*Q	Firm*Y*M

B: Include all Tweets

	Stock Return on Tweet day (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.137**	0.137**	0.113**	0.130**
	(2.61)	(2.46)	(2.58)	(2.92)
Neg. prop.	-0.081	-0.089	-0.152	-0.201*
	(-0.65)	(-0.74)	(-1.52)	(-2.20)
Total words	-0.000	-0.001	0.001	0.001
	(-0.24)	(-0.37)	(0.55)	(0.61)
Small minus big	0.340***	0.324***	0.338***	0.326***
	(4.37)	(4.33)	(4.53)	(4.49)
High minus low	-0.008	-0.001	-0.023	-0.014
	(-0.14)	(-0.02)	(-0.40)	(-0.24)
Market risk	0.766***	0.766***	0.764***	0.758***
premium	(19.29)	(20.64)	(20.72)	(21.07)
R square	0.171	0.206	0.186	0.247
N	118,842	118,744	126,465	125,881
Control	Yes	Yes	No	No
Fixed effect	Ind*Y*Q	Ind*Y*M	Firm*Y*Q	Firm*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 3 Predicting stock returns including days without Tweets

The following tables report the relation between daily stock returns and the fraction of positive words in Tweets, including days that CEOs do not post Tweets. The proportion of positive words is defined as the number of positive words divided by the total number of words in a Tweet, and it is equal to zero if there is no Tweet posted. The controls include the proportion of negative words, the total number of words, Fama-French factors, firm characteristics, and CEO characteristics. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We include industry or firm interacting with time fixed effects. Error is clustered at the firm and year level.

A: Exclude firm operations related Tweets

	Stock Return on Tweet day (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.182***	0.180**	0.120**	0.134***
	(3.19)	(2.82)	(2.86)	(3.33)
Neg. prop.	0.006	0.01	-0.126	-0.162
	(0.05)	(0.09)	(-1.22)	(-1.72)
Total words	0.002***	0.002***	0.001*	0.001**
	(3.93)	(4.13)	(2.06)	(2.39)
Small minus big	0.513***	0.511***	0.522***	0.520***
	(11.82)	(11.78)	(11.96)	(12.04)
High minus low	0.032	0.035	0.028	0.031
	(0.83)	(0.91)	(0.70)	(0.76)
Market risk	0.931***	0.928***	0.922***	0.918***
premium	(39.77)	(39.02)	(32.84)	(31.90)
R square	0.272	0.284	0.272	0.305
N	509,530	509,525	551,682	551,633
Controls	Yes	Yes	No	No
Fixed effect	Ind*Y*Q	Ind*Y*M	Firm*Y*Q	Firm*Y*M

B: Include all Tweets

	Stock Return on Tweet day (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.189***	0.187**	0.127***	0.142***
	(3.28)	(2.90)	(3.14)	(3.65)
Neg. prop.	-0.002	0.003	-0.136	-0.171
	(-0.02)	(0.03)	(-1.26)	(-1.74)
Total words	0.002***	0.002***	0.001*	0.001**
	(3.85)	(4.02)	(2.05)	(2.35)
Small minus big	0.513***	0.511***	0.522***	0.520***
	(11.83)	(11.80)	(11.98)	(12.06)
High minus low	0.032	0.035	0.028	0.031
	(0.83)	(0.91)	(0.69)	(0.77)
Market risk	0.931***	0.928***	0.922***	0.918***
premium	(39.80)	(39.06)	(32.86)	(31.93)
R square	0.272	0.284	0.272	0.304
N	511,041	511,036	553,322	553,273
Controls	Yes	Yes	No	No
Fixed effect	Ind*Y*Q	Ind*Y*M	Firm*Y*Q	Firm*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 4 Predicting subsequent stock returns

The following tables report the relation between future daily stock returns and the fraction of positive words in Tweets, including days without Tweets posted. The proportion of positive words is defined as the number of positive words divided by the total number of words in a Tweet; it is equal to zero if there is no Tweets posted. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We control for the Fama-French factors on the future days. We include firm interacting with time fixed effects. The error is clustered at the firm and year level. T-statistics are reported in the parenthesis.

A: Exclude firm operations related Tweets

	Future Daily Stock Return (%)			
	Day 1	Day 2	Day 3	Day 6
	(1)	(2)	(3)	(4)
Pos. prop	0.111***	0.093*	0.101**	-0.096**
	(4.70)	(1.87)	(2.23)	(-2.23)
Neg. prop.	-0.04	0.223	-0.058	-0.078
	(-0.43)	(1.77)	(-0.42)	(-0.63)
Total words	0.000	0.001	0.000	0.000
	(0.45)	(0.86)	(0.59)	(0.36)
Small minus big	0.575***	0.574***	0.573***	0.565***
	(13.25)	(13.33)	(13.61)	(13.3)
High minus low	0.017	0.015	0.012	0.014
	(0.40)	(0.36)	(0.28)	(0.32)
Market risk premium	0.959***	0.957***	0.958***	0.953***
	(25.87)	(26.27)	(26.3)	(26.18)
R square	0.327	0.327	0.328	0.324
N	551,412	551,182	550,971	550,356
controls	No	No	No	No
Fixed effect	Firm*Y*M	Firm*Y*M	Firm*Y*M	Firm*Y*M

B: Include all Tweets

	Future Daily Stock Return (%)			
	Day 1	Day 2	Day 3	Day 6
	(1)	(2)	(3)	(4)
Pos. prop	0.111***	0.093*	0.090*	-0.096**
	(3.74)	(2.03)	(1.96)	(-2.16)
Neg. prop.	-0.034	0.204	-0.045	-0.092
	(-0.36)	-1.56	(-0.33)	(-0.80)
Total words	0.000	0.001	0.000	0.000
	(0.45)	0.98	0.70	0.46
Small minus big	0.576***	0.574***	0.573***	0.565***
	13.25	13.34	13.58	13.3
High minus low	0.017	0.015	0.012	0.014
	0.40	0.37	0.28	0.31
Market risk premium	0.959***	0.957***	0.958***	0.953***
	25.83	26.27	26.24	26.16
R square	0.327	0.327	0.328	0.324
N	553,051	552,821	552,610	551,993
controls	No	No	No	No
Fixed effect	Firm*Y*M	Firm*Y*M	Firm*Y*M	Firm*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 5 Days prior to earnings announcements

The following tables report the relation between daily stock returns and the fraction of positive words in Tweets during two weeks prior to earnings announcements, including days that CEOs do not post Tweets. The proportion of positive words is defined as the number of positive words divided by the total number of words in a Tweet and it is equal to zero if there is no Tweet posted. The controls include the proportion of negative words, the total number of words, Fama-French factors, firm characteristics, and CEO characteristics. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We include industry or firm interacting with time fixed effects. The error is clustered at the firm level.

A: Exclude firm operations related Tweets

	Stock Return on Tweet day (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.452***	0.440***	0.411***	0.408**
	(2.74)	(2.60)	(2.63)	(2.35)
Neg. prop.	-0.355	-0.273	-0.568*	-0.45
	(-1.19)	(-0.93)	(-1.81)	(-1.42)
Total words	0.001	0.001	0.002	0.002
	(0.77)	(0.72)	(0.72)	(0.78)
Small minus big	0.559***	0.563***	0.574***	0.574***
	(11.58)	(11.29)	(11.99)	(11.82)
High minus low	0.073*	0.073	0.074*	0.07
	(1.69)	(1.65)	(1.70)	(1.59)
Market risk	0.942***	0.940***	0.931***	0.930***
premium	(38.75)	(38.91)	(38.39)	(38.64)
R square	0.294	0.316	0.337	0.372
N	74,892	74,826	79,716	79,488
Controls	Yes	Yes	No	No
Fixed effect	Ind*Y*Q	Ind*Y*M	Firm*Y*Q	Firm*Y*M

B: Include all Tweets

	Stock Return on Tweet day (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.431***	0.422**	0.386**	0.392**
	(2.67)	(2.55)	(2.49)	(2.28)
Neg. prop.	-0.346	-0.257	-0.563*	-0.444
	(-1.16)	(-0.88)	(-1.80)	(-1.42)
Total words	0.002	0.002	0.002	0.002
	(0.84)	(0.81)	(0.80)	(0.81)
Small minus big	0.558***	0.562***	0.573***	0.574***
	(11.54)	(11.26)	(11.97)	(11.81)
High minus low	0.072*	0.073	0.074*	0.07
	(1.66)	(1.63)	(1.69)	(1.58)
Market risk premium	0.941***	0.940***	0.930***	0.929***
	(38.72)	(38.85)	(38.34)	(38.55)
R square	0.294	0.316	0.336	0.371
N	75,121	75,055	79,971	79,744
Controls	Yes	Yes	No	No
Fixed effect	Ind*Y*Q	Ind*Y*M	Firm*Y*Q	Firm*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 6 Predicting stock return on earnings announcements

The following tables report the relation between stock returns on earnings announcements and the fraction of positive words in Tweets during two weeks prior to earnings announcements, including days without Tweets posted. The proportion of positive words is defined as the number of positive words divided by the total number of words in a Tweet; it is equal to zero if there is no Tweets posted. The dependent variable is daily stock returns on earnings announcements. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. The controls include firm characteristics and CEO features. We include industry interacting with time fixed effects. The error is clustered at the firm level. T-statistics are reported in the parenthesis.

A: Exclude firm operations related Tweets

	Daily stock return on earnings announcements (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.861***	0.776***	0.780***	0.714***
	(2.95)	(2.79)	(2.76)	(2.62)
Neg. prop.	-1.273	-1.277	-1.281	-1.176
	(-1.20)	(-1.20)	(-1.28)	(-1.15)
Total words	0.013**	0.011**	0.014***	0.013**
	(2.41)	(2.06)	(2.78)	(2.49)
Small minus big			0.668***	0.672***
			(5.64)	(5.67)
High minus low			-0.153	-0.081
			(-1.57)	(-0.74)
Market risk premium			0.927***	0.936***
			(17.98)	(16.98)
R square	0.379	0.467	0.453	0.53
N	64,671	64,604	64,671	64,604
controls	Yes	Yes	Yes	Yes
Fixed effect	Ind*Y*Q	Ind*Y*M	Ind*Y*Q	Ind*Y*M

B: Include all Tweets

	Daily stock return on earnings announcements (%)			
	(1)	(2)	(3)	(4)
Pos. prop	0.850***	0.757***	0.768***	0.696**
	(2.91)	(2.73)	(2.72)	(2.56)
Neg. prop.	-1.415	-1.411	-1.407	-1.293
	(-1.36)	(-1.35)	(-1.43)	(-1.29)
Total words	0.013**	0.011**	0.015***	0.013**
	(2.42)	(2.07)	(2.81)	(2.52)
Small minus big			0.669***	0.673***
			(5.65)	(5.68)
High minus low			-0.153	-0.082
			(-1.57)	(-0.75)
Market risk premium			0.928***	0.937***
			(18.02)	(17.02)
R square	0.379	0.467	0.453	0.53
N	64,775	64,708	64,775	64,708
controls	Yes	Yes	Yes	Yes
Fixed effect	Ind*Y*Q	Ind*Y*M	Ind*Y*Q	Ind*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 7 Predicting stock return on earnings announcements using cumulative positive words
The following tables report the relation between daily stock returns on earnings announcements and the cumulative proportion of positive words in Tweets prior to earnings announcements. The cumulative measurements of Tweets are counted starting 14 days and ending 1 day prior to earnings announcements. The independent variable of interest is the cumulative proportion of positive words, and the dependent variable is daily stock returns on earnings announcements. The controls include firm characteristics and CEO features. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We include industry and time fixed effects. Error is clustered at the firm level. T-statistics are reported in the parenthesis.

A: Exclude firm operations related Tweets

	Daily stock return on earnings announcements (%)			
	(1)	(2)	(3)	(4)
Cum. pos. prop.	1.614*	1.749*	1.651*	1.762*
	(2.03)	(2.16)	(1.89)	(2.04)
Cum. neg. prop.	-6.773	-6.548	-6.523	-6.428
	(-1.57)	(-1.50)	(-1.58)	(-1.56)
Cum. Tweets	0.029	0.029	0.026	0.027
	(1.04)	(1.07)	(0.99)	(1.05)
Small minus big			0.553***	0.549***
			-4.37	(4.36)
High minus low			-0.179	-0.167
			(-1.51)	(-1.38)
Market risk premium			0.861***	0.858***
			(13.62)	(12.78)
R square	0.021	0.023	0.118	0.119
N	4,241	4,241	4,241	4,241
Control	Yes	Yes	Yes	Yes
Fixed effect	Ind, Y	Ind, Y ,Q	Ind, Y	Ind, Y ,Q

B: Include all Tweets

	Daily stock return on earnings announcements (%)			
	(1)	(2)	(3)	(4)
Cum. pos. prop.	1.659*	1.798**	1.693*	1.806*
	(2.12)	(2.25)	(2.02)	(2.16)
Cum. neg. prop.	-6.97	-6.757	-6.77	-6.688
	(-1.58)	(-1.52)	(-1.61)	(-1.60)
Cum. Tweets	0.029	0.029	0.026	0.026
	(1.08)	(1.11)	(1.01)	(1.07)
Small minus big			0.553***	0.548***
			(4.37)	(4.36)
High minus low			-0.179	-0.167
			(-1.51)	(-1.38)
Market risk premium			0.862***	0.858***
			(13.6)	(12.76)
R square	0.021	0.023	0.118	0.119
N	4,241	4,241	4,241	4,241
Control	Yes	Yes	Yes	Yes
Fixed effect	Ind, Y	Ind, Y ,Q	Ind, Y	Ind, Y ,Q

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 8 Portfolio returns: 14 days before earnings announcement

The following tables report Fama-French factor alphas on daily portfolios that buy stocks with a high proportion of positive words in their CEOs' Tweets and sell stocks with a low fraction of positive words during the 14 days prior to earnings announcements. The sample contains 988 days that have at least two Tweets posted, including at least one Tweet with positive words. We sort the Tweets based on the proportion of positive words on Tweet days (Day 0) and rebalance the portfolios daily. We long stocks that rank in the top 10th percentile of the proportion of positive words and short the stocks in the bottom 10th percentile, using either equal- or value-weighting schemes across firms. The holding period is a day in Panel B-C and a week in Panel D. The following tables report alphas and factor loadings by regressing daily stock returns on risk factors.

A: Summary statistics

	Sort proportion of positive words on Day 0	
	Top 10	Bottom 10
	(1)	(2)
Prop. Pos	0.194	0.000
	(0.188)	(0.000)
Num. of Tweets each day	1.235	1.235
	(0.519)	(0.519)
Day 1 return (%)	0.189	0.045
	(2.650)	(2.702)
Portfolio Day 1	0.210	0.056
return (%)	(2.464)	(2.492)
Num of Tweets in portfolio	1,220	1,200

B: Long stocks that rank in the top 10 percentiles of the prop. pos. words (equal-weighted)

	Stock return on Day 1 (%)		
	Positive words	Lack positive words	Positive – lack positive
	(1)	(2)	(3)
Alpha	0.128**	-0.044	0.172***
	(2.60)	(-0.86)	(3.50)
Small minus big	0.588***	0.452***	0.136
	(3.80)	(3.53)	(0.72)
High minus low	-0.304	-0.153	-0.151
	(-1.67)	(-1.35)	(-0.81)
Market risk premium	0.938***	1.113***	-0.175*
	(17.35)	(14.32)	(-2.15)

Electronic copy available at: https://ssrn.com/abstract=3293134

Cont. of Table 8

C: Long stocks that rank in the top 10 percentiles of the prop. pos. words (value-weighted)

	Stock return on Day 1 (%)		
	Positive words	Lack positive words	Positive – lack positive
	(1)	(2)	(3)
Alpha	0.133***	-0.063	0.196***
	(3.42)	(-1.34)	(3.57)
Small minus big	0.518***	0.410***	0.108
	(3.31)	(3.58)	(0.55)
High minus low	-0.313*	-0.157	-0.155
	(-1.96)	(-1.50)	(-0.92)
Market risk premium	0.929***	1.117***	-0.188**
	(18.41)	(17.41)	(-2.79)

D: Long stocks that rank in the top 10 percentiles and hold for a week (equal-weighted)

	Stock return in week 1 (%)		
	Positive words	Lack positive words	Positive – lack positive
	(1)	(2)	(3)
Alpha	0.125***	0.016	0.108**
	(2.81)	(0.38)	(2.27)
Small minus big	0.501***	0.465***	0.037
	(5.45)	(5.25)	(0.37)
High minus low	-0.135	0.003	-0.138
	(-1.45)	(0.03)	(-1.37)
Market risk premium	0.990***	1.068***	-0.078
	(19.45)	(21.78)	(-1.43)

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 9 Predicting earnings surprise using the proportion of positive words

The following tables report the relationship between earnings surprise and the cumulative proportion of positive words in Tweets prior to earnings announcements. The cumulative measurements of Tweets are counted starting 14 days and ending 1 day prior to earnings announcements. The independent variable of interest is the cumulative proportion of positive words, and the dependent variable is standardized unexpected earnings. Standardized unexpected earnings are measured as the difference between actual quarterly earnings and the mean of analysts' forecasts (earnings surprise), divided by the standard deviation of earnings surprise. The controls include the cumulative proportion of negative words, the cumulative number of Tweets, firm characteristics, and CEO characteristics. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We include industry or firm and time fixed effects. Error is clustered at the firm level. T-statistics are reported in the parenthesis.

A: Exclude firm operations related Tweets

	Standardized Unexpected Earnings			
	(1)	(2)	(3)	(4)
Cum. pos. prop.	0.797*	0.784*	0.938*	0.925*
	(1.83)	(1.83)	(1.94)	(1.96)
Cum. neg. prop.	-0.996	-0.951	-1.182	-1.130
	(-0.80)	(-0.76)	(-0.86)	(-0.81)
Cum. Tweets	-0.029	-0.030	-0.033	-0.035
	(-0.74)	(-0.77)	(-0.93)	(-0.99)
R square	0.047	0.048	0.111	0.111
N	4,121	4,121	4,114	4,114
Control	Yes	Yes	Yes	Yes
Fixed effect	Ind, Y	Ind, Y ,Q	Firm, Y	Firm, Y, Q

B: Include all Tweets

	Standardized Unexpected Earnings			
	(1)	(2)	(3)	(4)
Cum. pos. prop.	0.813*	0.802*	0.941*	0.930*
	(1.85)	(1.84)	(1.93)	(1.94)
Cum. neg. prop.	-0.922	-0.877	-1.124	-1.070
	(-0.75)	(-0.71)	(-0.81)	(-0.77)
Cum. Tweets	-0.030	-0.032	-0.035	-0.037
	(-0.78)	(-0.81)	(-0.96)	(-1.02)
R square	0.047	0.048	0.111	0.111
N	4,121	4,121	4,114	4,114
Control	Yes	Yes	Yes	Yes
Fixed effect	Ind, Y	Ind, Y ,Q	Firm, Y	Firm, Y, Q

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 10 Predicting earnings surprise using the number of positive words

The following tables report the relationship between earnings surprise and the cumulative number of positive words in Tweets prior to earnings announcements. The cumulative measurements of Tweets are counted starting 14 days and ending 1 day prior to earnings announcements. The independent variable of interest is the cumulative number of positive words, and the dependent variable is standardized unexpected earnings. Standardized unexpected earnings are measured as the difference between actual quarterly earnings and the mean of analysts' forecasts (earnings surprise), divided by the standard deviation of earnings surprise. The controls include the cumulative number of negative words, the cumulative number of Tweets, firm characteristics, and CEO characteristics. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets We include industry or firm and time fixed effects. Error is clustered at the firm level. T-statistics are reported in the parenthesis.

A: Exclude firm operations related Tweets

	Standardized Unexpected Earnings			
	(1)	(2)	(3)	(4)
Cum. pos. num.	0.107**	0.103**	0.112**	0.109**
	(2.49)	(2.41)	(2.24)	(2.19)
Cum. neg. num.	0.031	0.034	0.007	0.012
	(0.44)	(0.48)	(0.10)	(0.15)
Cum. Tweets	-0.079*	-0.079*	-0.076*	-0.077*
	(-1.85)	(-1.83)	(-1.93)	(-1.95)
R square	0.047	0.048	0.111	0.111
N	4,121	4,121	4,114	4,114
Control	Yes	Yes	Yes	Yes
Fixed effect	Ind, Y	Ind, Y ,Q	Firm, Y	Firm, Y, Q

Panel B: Include all Tweets

	Standardized Unexpected Earnings			
	(1)	(2)	(3)	(4)
Cum. pos. num.	0.107**	0.104**	0.112**	0.108**
	(2.52)	(2.44)	(2.24)	(2.19)
Cum. neg. num.	0.034	0.037	0.009	0.014
	(0.48)	(0.52)	(0.13)	(0.18)
Cum. Tweets	-0.079*	-0.079*	-0.077*	-0.077*
	(-1.87)	(-1.86)	(-1.95)	(-1.97)
R square	0.047	0.048	0.111	0.111
N	4,121	4,121	4,114	4,114
Control	Yes	Yes	Yes	Yes
Fixed effect	Ind, Y	Ind, Y ,Q	Firm, Y	Firm, Y, Q

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 11 Predicting earnings surprise using alternative timeframes

The following tables report the relationship between earnings surprise and the cumulative proportion or number of positive words in Tweets prior to earnings announcements, using alternative timeframes. In columns 1 and 2, the cumulative measurements of Tweets are counted starting 7 days and ending 1 day prior to earnings announcements. In columns 3 and 4, the cumulative measurements are counted starting 28 days and ending 1 day prior to earnings announcements. The independent variable of interest is the cumulative proportion of positive words in Panel A, and the cumulative number of positive words in Panel B. The dependent variables are standardized unexpected earnings, which are measured as the difference between actual quarterly earnings and the mean of analysts' forecasts (earnings surprise), divided by the standard deviation of earnings surprise. The controls include the cumulative proportion or number of negative words, the cumulative number of Tweets, firm characteristics, and CEO characteristics. We include firm and time fixed effects. Error is clustered at the firm level. T-statistics are reported in the parenthesis.

A: Proportion of positive words

	Standardized Unexpected Earnings			
	One week		Four weeks	
	(1)	(2)	(3)	(4)
Cum. pos. prop.	1.045*	1.039*	0.644*	0.631*
	(1.74)	(1.76)	(1.73)	(1.72)
Cum. neg. prop.	-1.314	-1.281	-0.044	-0.029
	(-0.63)	(-0.61)	(-0.05)	(-0.03)
Cum. Tweets	-0.047	-0.050	-0.019	-0.020
	(-1.05)	(-1.10)	(-0.75)	(-0.79)
R square	0.111	0.111	0.112	0.112
N	4,106	4,106	4,128	4,128
Control	Yes	Yes	Yes	Yes
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

B: Number of positive words

	Standardized Unexpected Earnings			
	One week		Four weeks	
	(1)	(2)	(3)	(4)
Cum. pos. num.	0.111*	0.107*	0.067*	0.065*
	(1.80)	(1.76)	(1.81)	(1.77)
Cum. neg. num.	0.070	0.073	0.049	0.050
	(0.68)	(0.68)	(-0.90)	(0.91)
Cum. Tweets	-0.094*	-0.094*	-0.041	-0.041
	(-1.83)	(-1.83)	(-1.52)	(-1.53)
R square	0.111	0.111	0.112	0.112
N	4,106	4,106	4,128	4,128
Control	Yes	Yes	Yes	Yes
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 12 Predicting earnings per share using positive words

The following tables report the relationship between earnings per shares and the cumulative positive words in Tweets. The cumulative measurements of Tweets are counted starting 14 days and ending 1 day prior to the end of each quarter. The independent variable of interest is the cumulative proportion of positive words, and dependent variables are earnings per share at the end of each quarter. Earnings per share are measured as the diluted EPS dividend by lagged stock price in columns 1-2, basic EPS dividend by lagged stock price in columns 3-4, income before extraordinary items divided by lagged capitalization in columns 5-6, income before extraordinary items adjusted for common stock equivalents divided by lagged capitalization in columns 7-8, and income before extraordinary items available for common divided by lagged capitalization in columns 9-10. The controls include the cumulative proportion of negative words, the cumulative number of Tweets, and firm and CEO characteristics. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We include firm and time fixed effects. Error is clustered at the firm level.

A: Exclude firm operations related Tweets

	Diluted EPS /price		Basic EPS /price		Income before /cap.		Income before adj./cap.		Income before common /cap.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cum. pos.	0.034**	0.033**	0.033**	0.033**	0.033**	0.033**	0.034**	0.034**	0.034**	0.034**
prop.	(2.21)	(2.19)	(2.17)	(2.15)	(2.19)	(2.17)	(2.21)	(2.20)	(2.20)	(2.18)
Cum.	-0.023	-0.022	-0.024	-0.023	-0.023	-0.022	-0.023	-0.022	-0.023	-0.022
neg. prop.	(-0.90)	(-0.84)	(-0.93)	(-0.87)	(-0.91)	(-0.85)	(-0.90)	(-0.84)	(-0.90)	(-0.84)
Cum.	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Tweets	(-0.57)	(-0.54)	(-0.49)	(-0.45)	(-0.50)	(-0.47)	(-0.53)	(-0.49)	(-0.54)	(-0.50)
R square	0.227	0.229	0.225	0.228	0.23	0.232	0.228	0.231	0.228	0.23
N	4,833	4,833	4,833	4,833	4,831	4,831	4,831	4,831	4,831	4,831
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

B: Include all Tweets

	Diluted EPS /price		Basic EPS /price		Income before /cap.		Income before adj./cap.		Income before common /cap.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cum. pos.	0.033**	0.033**	0.033**	0.033**	0.033**	0.033**	0.034**	0.034**	0.033**	0.033**
prop.	(2.18)	(2.17)	(2.14)	(2.13)	(2.16)	(2.14)	(2.19)	(2.17)	(2.17)	(2.16)
Cum.	-0.02	-0.019	-0.02	-0.02	-0.02	-0.019	-0.02	-0.019	-0.02	-0.019
neg. prop.	(-0.76)	(-0.71)	(-0.79)	(-0.73)	(-0.77)	(-0.72)	(-0.76)	(-0.71)	(-0.76)	(-0.71)
Cum.	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Tweets	(-0.55)	(-0.52)	(-0.47)	(-0.44)	(-0.48)	(-0.45)	(-0.50)	(-0.47)	(-0.51)	(-0.48)
R square	0.227	0.229	0.225	0.228	0.23	0.232	0.228	0.231	0.228	0.23
N	4,833	4,833	4,833	4,833	4,831	4,831	4,831	4,831	4,831	4,831
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 13 Predicting earnings using positive words

The following tables report the relationship between normalized earnings and the cumulative positive words in Tweets. The cumulative measurements of Tweets are counted starting 14 days and ending 1 day prior to the end of each quarter. The independent variable of interest is the cumulative proportion of positive words, and dependent variables are earnings at the end of each quarter. Earnings are measured as 4 times the net income divided by lagged assets in columns 1-2, income before extraordinary items divided by lagged assets in columns 3-4, income before extraordinary items adjusted for common stock equivalents divided by lagged assets in columns 5-6, income before extraordinary items available for common divided by lagged assets in columns 8-9, and cash flow (sum of net income and depreciation expenses) divided by lagged assets in columns 9-10. The controls include the cumulative proportion of negative words, the cumulative number of Tweets, and firm and CEO characteristics. Panel A excludes Tweets related to firm operations and Panel B contains all of the Tweets. We include firm and time fixed effects. Error is clustered at the firm level.

A: Exclude firm operations related Tweets

	Net income /assets		Income before /assets		Income before adj. /assets		Income before common / assets		Cash flow /assets	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cum. pos. prop.	0.091** (2.28)	0.090** (2.26)	0.082** (2.14)	0.082** (2.12)	0.082** (2.15)	0.082** (2.12)	0.082** (2.15)	0.082** (2.13)	0.023** (2.33)	0.023** (2.31)
Cum. neg. prop.	0.022 (0.34)	0.023 (0.35)	0.011 (0.18)	0.012 (0.20)	0.011 (0.17)	0.012 (0.20)	0.011 (0.17)	0.012 (0.20)	-0.005 (-0.32)	-0.005 (-0.30)
Cum. Tweets	0.000 (-0.12)	0.000 (-0.10)	0.000 (-0.17)	0.000 (-0.15)	0.000 (-0.17)	0.000 (-0.16)	0.000 (-0.17)	0.000 (-0.16)	0.000 (-0.07)	0.000 (-0.06)
R square	0.321	0.322	0.349	0.351	0.349	0.351	0.349	0.351	0.355	0.356
N	4,859	4,859	4,859	4,859	4,859	4,859	4,859	4,859	4,714	4,714
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

B: Include all Tweets

	Net income /assets		Income before /assets		Income before adj. /assets		Income before common / assets		Cash flow /assets	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cum. pos. prop.	0.085** (2.16)	0.085** (2.15)	0.076** (2.01)	0.076** (2.00)	0.076** (2.01)	0.076** (2.00)	0.076** (2.02)	0.076** (2.00)	0.022** (2.23)	0.022** (2.21)
Cum. neg. prop.	0.037 (0.54)	0.038 (0.54)	0.028 (0.44)	0.029 (0.44)	0.028 (0.43)	0.029 (0.44)	0.028 (0.43)	0.029 (0.44)	-0.002 (-0.091)	-0.001 (-0.082)
Cum. Tweets	0.000 (-0.14)	0.000 (-0.13)	0.000 (-0.19)	0.000 (-0.18)	0.000 (-0.20)	0.000 (-0.18)	0.000 (-0.200)	0.000 (-0.18)	0.000 (-0.09)	0.000 (-0.08)
R square	0.321	0.322	0.349	0.35	0.349	0.351	0.349	0.351	0.355	0.356
N	4,859	4,859	4,859	4,859	4,859	4,859	4,859	4,859	4,714	4,714
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

Electronic copy available at: https://ssrn.com/abstract=3293134

Table 14 Comparing Tweets around option exercises and stock sales

The following tables report characteristics of Tweets posted during 14 days before and 14 days after CEO option exercises in Panel A and stock sales in Panel B. The characteristics of Tweets include the proportion of positive and negative words, and the number of words. In panel A, the independent variable of interest is the before CEO option exercise indicator, which takes on a value of 1 if it is posted within 14 days prior to option exercises, and it is equal to 0 if a Tweet is posted within 14 days after option exercises. In panel B, the independent variable of interest is the before CEO stock sale date indicator, which takes on a value of 1 if a Tweet is posted within 14 days prior to stock sales, and it is equal to 0 if a Tweet is posted within 14 days after stock sales. The dependent variable is the proportion of positive words in a Tweet in columns 1-2, the proportion of negative words in a Tweet in columns 3-4, and the number of word in a Tweet in columns 5-6. We include firm and time fixed effects. Error is clustered at the firm level. T-statistics are reported in the parenthesis.

A: Days around option exercises

	Proportion of Positive Words (%)		Proportion of Negative Words (%)		Total number of Words	
	(1)	(2)	(3)	(4)	(5)	(6)
I (before CEO option exercises)	0.124*	0.123*	0.024	0.024	0.332***	0.332***
	(1.73)	(1.72)	(0.90)	(0.86)	(2.69)	(2.64)
R square	0.19	0.19	0.082	0.083	0.448	0.45
N	30,933	30,933	30,933	30,933	30,965	30,965
Control	No	No	No	No	No	No
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

B: Days around stock sales

	Proportion of Positive Words (%)		Proportion of Negative Words (%)		Total number of Words	
	(1)	(2)	(3)	(4)	(5)	(6)
I (before CEO stock sales)	0.132**	0.130*	0.020	0.019	0.207*	0.206*
	(2.01)	(1.95)	(0.77)	(0.74)	(1.71)	(1.69)
R square	0.154	0.154	0.073	0.074	0.384	0.392
N	55,330	55,330	55,330	55,330	55,391	55,392
Control	No	No	No	No	No	No
Fixed effect	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q	Firm, Y	Firm, Y, Q

Electronic copy available at: https://ssrn.com/abstract=3293134

APPENDIX

Table A.1 Predicting stock returns controlling for lagged stock returns

The following tables report the relation between daily stock returns and the fraction of positive words in Tweets, controlling for lagged daily stock returns. Columns 1-2 replicate Table 2, columns 3-4 replicate Table 3, and columns 5-6 replicate Table 5.

A: Exclude firm operations related Tweets

	Stock Return on Tweet day (%)					
	Only Days with Tweets		Including days without Tweets		Prior to earnings release	
	(1)	(2)	(3)	(4)	(5)	(6)
Pos. prop	0.128**	0.118**	0.173**	0.127***	0.429**	0.412**
	(2.35)	(2.59)	(2.89)	(3.11)	(2.49)	(2.38)
Neg. prop.	-0.077	-0.194*	0.021	-0.154	-0.265	-0.454
	(-0.63)	(-2.11)	(0.19)	(-1.64)	(-0.90)	(-1.43)
Total words	-0.000	0.001	0.002***	0.001**	0.001	0.002
	(-0.35)	(0.70)	(4.37)	(2.41)	(0.75)	(0.77)
Small minus big	0.323***	0.324***	0.501***	0.513***	0.562***	0.574***
	(4.36)	(4.47)	(11.93)	(12.19)	(11.31)	(11.81)
High minus low	-0.000	-0.013	0.03	0.026	0.072	0.071
	(-0.01)	(-0.23)	(0.81)	(0.69)	(1.61)	(1.60)
Market risk	0.759***	0.755***	0.920***	0.913***	0.939***	0.930***
premium	(20.9)	(21.54)	(40.56)	(32.92)	(38.60)	(38.75)
Lagged stock	0.032***	0.014**	0.087***	0.063***	0.037***	-0.014*
return	(4.12)	(2.35)	(17.06)	(22.78)	(3.67)	(-1.87)
R square	0.206	0.247	0.291	0.309	0.317	0.372
N	117,210	124,168	509,389	551,323	74,825	79,481
Controls	Yes	No	Yes	No	Yes	No
Fixed effect	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M

B: Include all Tweets

	Stock Return on Tweet day (%)					
	Only Days with Tweets		Including days without Tweets		Prior to earnings release	
	(1)	(2)	(3)	(4)	(5)	(6)
Pos. prop	0.135**	0.125**	0.180**	0.135***	0.410**	0.396**
	(2.46)	(2.75)	(2.98)	(3.46)	(2.43)	(2.31)
Neg. prop.	-0.083	-0.203*	0.014	-0.163	-0.249	-0.448
	(-0.70)	(-2.19)	(0.13)	(-1.67)	(-0.85)	(-1.43)
Total words	-0.001	0.001	0.002***	0.001**	0.002	0.002
	(-0.35)	(0.58)	(4.18)	(2.37)	(0.84)	(0.80)
Small minus big	0.324***	0.325***	0.501***	0.513***	0.561***	0.573***
	(4.40)	(4.52)	(11.94)	(12.21)	(11.28)	(11.79)
High minus low	0.000	-0.012	0.03	0.026	0.071	0.071
	(0.00)	(-0.21)	(0.81)	(0.70)	(1.59)	(1.59)
Market risk	0.761***	0.757***	0.920***	0.913***	0.939***	0.929***
premium	(20.85)	(21.51)	(40.60)	(32.95)	(38.53)	(38.65)
Lagged stock	0.032***	0.014**	0.087***	0.063***	0.036***	-0.014*
return	(4.15)	(2.30)	(17.10)	(22.74)	(3.61)	(-1.86)
R square	0.206	0.247	0.291	0.308	0.317	0.371
N	118,720	125,797	510,899	552,962	75,054	79,737
Controls	Yes	No	Yes	No	Yes	No
Fixed effect	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134

Table A.2 Predicting stock returns controlling for Fama-French 5 factors and Momentum
The following tables report the relation between daily stock returns and the fraction of positive words in Tweets, controlling for Robust Minus Weak, Conservative Minus Aggressive and momentum. Columns 1-2 replicate Table 2, columns 3-4 replicate Table 3, and columns 5-6 replicate Table 5.

A: Exclude firm operations related Tweets

	Stock Return on Tweet day (%)					
	Only Days with Tweets		Including days without Tweets		Prior to earnings release	
	(1)	(2)	(3)	(4)	(5)	(6)
Pos. prop	0.131**	0.124**	0.181**	0.135***	0.439**	0.407**
	(2.24)	(2.83)	(2.81)	(3.35)	(2.59)	(2.34)
Neg. prop.	-0.081	-0.192*	0.011	-0.162	-0.273	-0.448
	(-0.68)	(-2.17)	(0.09)	(-1.73)	(-0.92)	(-1.40)
Total words	-0.001	0.001	0.002***	0.001**	0.001	0.002
	(-0.29)	(0.72)	(3.09)	(2.35)	(0.70)	(0.78)
Small minus big	0.328***	0.329***	0.509***	0.515***	0.544***	0.555***
	(3.97)	(4.02)	(12.12)	(12.45)	(10.81)	(11.36)
High minus low	0.028	0.005	0.036	0.024	0.068	0.059
	(0.54)	(0.08)	(1.22)	(0.82)	(1.52)	(1.38)
Market risk	0.761***	0.755***	0.919***	0.909***	0.923***	0.913***
premium	(21.07)	(20.96)	(37.68)	(29.86)	(39.98)	(40.13)
Robust minus	-0.005	-0.002	-0.02	-0.033	-0.118*	-0.120*
weak	(-0.04)	(-0.02)	(-0.40)	(-0.60)	(-1.89)	(-1.91)
Conservative	-0.045	-0.02	-0.07	-0.058	-0.092	-0.08
minus aggressive	(-0.31)	(-0.14)	(-1.62)	(-1.38)	(-1.22)	(-1.07)
Momentum	0.024	0.021	-0.015	-0.019	-0.009	-0.012
	(0.99)	(0.75)	(-0.77)	(-1.11)	(-0.41)	(-0.57)
R square	0.205	0.247	0.284	0.305	0.317	0.373
N	117,233	124,251	509,525	551,633	74,826	79,488
Controls	Yes	No	Yes	No	Yes	No
Fixed effect	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134

B: Include all Tweets

	Stock Return on Tweet day (%)					
	Only Days with Tweets		Including days without Tweets		Prior to earnings release	
	(1)	(2)	(3)	(4)	(5)	(6)
Pos. prop	0.138**	0.131**	0.188**	0.142***	0.420**	0.391**
	(2.35)	(2.98)	(2.90)	(3.66)	(2.53)	(2.27)
Neg. prop.	-0.088	-0.201**	0.004	-0.171	-0.257	-0.442
	(-0.75)	(-2.25)	(0.03)	(-1.75)	(-0.87)	(-1.40)
Total words	-0.001	0.001	0.002**	0.001**	0.001	0.002
	(-0.29)	(0.60)	(2.92)	(2.31)	(0.79)	(0.81)
Small minus big	0.328***	0.329***	0.509***	0.515***	0.544***	0.554***
	(4.01)	(4.06)	(12.13)	(12.46)	(10.77)	(11.34)
High minus low	0.026	0.003	0.036	0.023	0.067	0.059
	(0.48)	(0.06)	(1.19)	(0.80)	(1.49)	(1.37)
Market risk	0.763***	0.757***	0.919***	0.909***	0.922***	0.912***
premium	(21.26)	(21.29)	(37.7)	(29.87)	(39.89)	(39.98)
Robust minus	-0.005	-0.003	-0.02	-0.033	-0.118*	-0.120*
weak	(-0.04)	(-0.03)	(-0.40)	(-0.60)	(-1.89)	(-1.90)
Conservative	-0.039	-0.015	-0.069	-0.056	-0.092	-0.08
minus aggressive	(-0.27)	(-0.11)	(-1.58)	(-1.35)	(-1.22)	(-1.06)
Momentum	0.022	0.02	-0.016	-0.02	-0.01	-0.013
	(0.89)	(0.70)	(-0.79)	(-1.13)	(-0.44)	(-0.60)
R square	0.206	0.247	0.284	0.305	0.316	0.372
N	118,744	125,881	511,036	553,273	75,055	79,744
Controls	Yes	No	Yes	No	Yes	No
Fixed effect	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M	Ind*Y*M	Firm*Y*M

Electronic copy available at: https://ssrn.com/abstract=3293134